

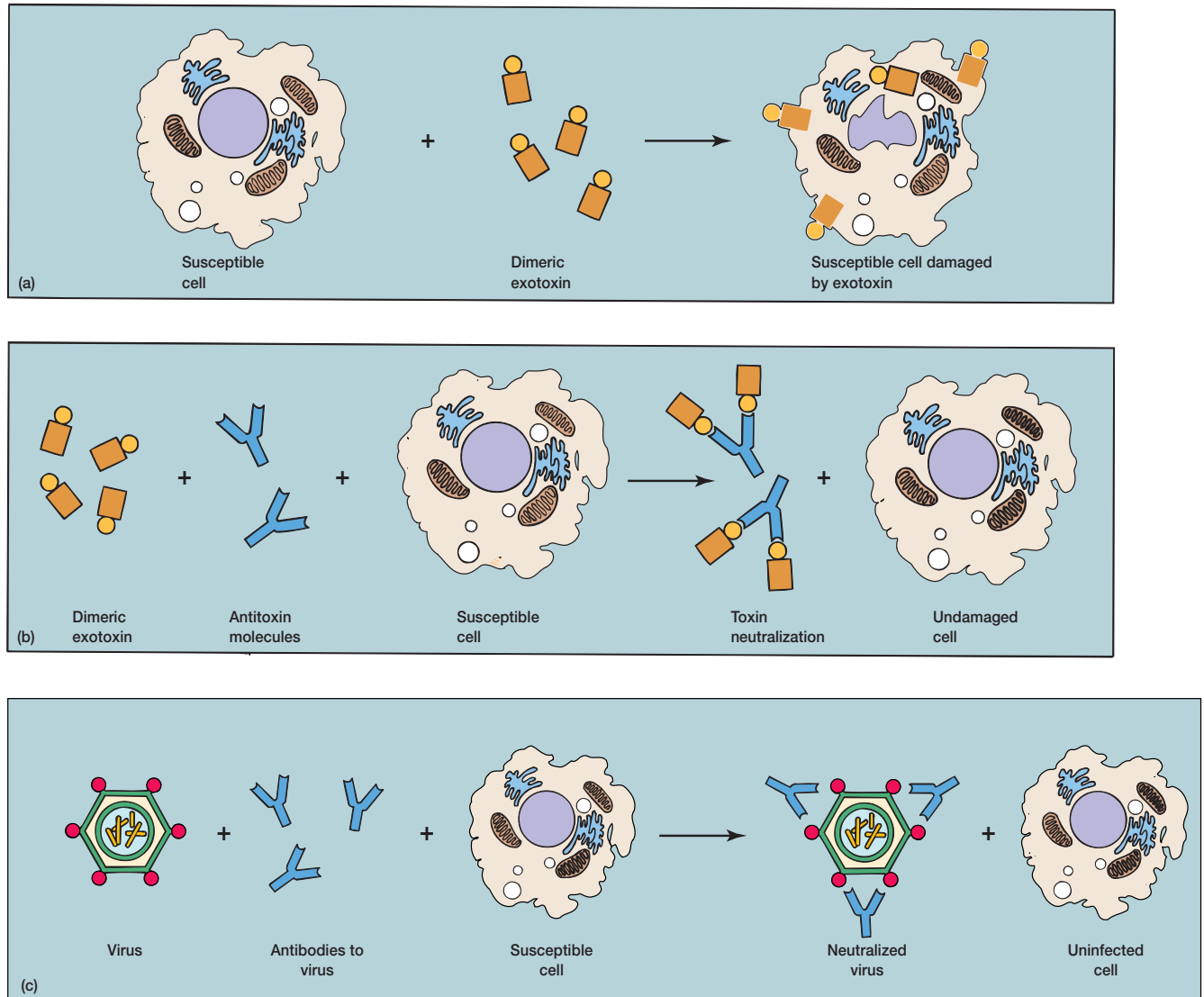


Figure 32.29 Neutralization Reactions. (a) Effects of a dimeric exotoxin on a susceptible cell. (b) Neutralization of the toxin by antitoxin. (c) In viral neutralization the specific antibodies neutralize the virus and prevent it from attaching to the susceptible cells.

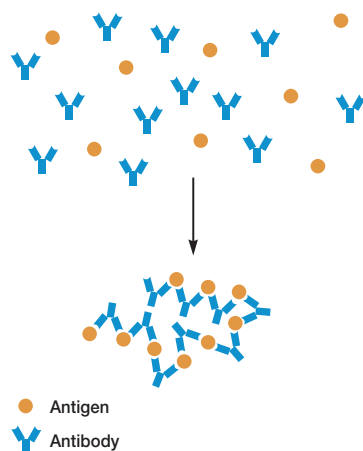


Figure 32.30 Immune Complex Formation. Antibodies cross-link antigens forming an aggregate of antibody and antigen termed an immune complex.

1. How does toxin neutralization occur? Viral neutralization?
2. Describe the role of IgE in resistance to parasitic infections.
3. How does adherence inhibition occur?
4. Describe an immune complex. What two types are formed?

1. How is the classical complement pathway activated?
2. What effect does the membrane attack complex have on eucaryotic cells? Prokaryotic cells?

32.7 The Classical Complement Pathway

Activation of the **classical complement pathway** requires initiation by the interaction of antibodies with an antigen that is usually cell bound (**figure 32.31**). The order of effectiveness in activating complement is as follows: IgM > IgG3 > IgG1 > IgG2. However, some microbial products (lipid A of endotoxin and staphylococcal protein A) or plasmin (a proteolytic enzyme that dissolves the fibrin of blood clots) may activate C1 directly without antibody participation. Following binding of antigen to antibody, the C1 complement component, which is composed of three proteins (q, r, and s), attaches to the Fc portion of the antibody molecule through its C1q subcomponent. In the presence of calcium ions, a trimolecular complex (C1qrs · Ag · Ab) that has esterase activity is rapidly formed. The activated C1's subcomponent attacks and cleaves its natural substrates in serum (C2 and C4). This leads to binding of a portion of each molecule (C2b and C4b) to the antigen-antibody-complement complex with the release of small C4a and C2a fragments. (The released C2 fragment traditionally has been called C2a. For consistency, this text will label all larger complement fragments that are bound to the target cell as b fragments.) With the binding of C2b to C4b, an enzyme with trypsinlike proteolytic activity is generated. The natural substrate for this enzyme is C3; thus it is termed a C3 convertase. Through the activity of C4b2b (the bar indicates an active enzyme complex), C3 is cleaved into a bound subcomponent C3b and a C3a soluble component. C3b then absorbs to bound C4b2b, forming the complex C4b2b3b, which acts as a C5 convertase and cleaves C5 into fragments C5a and C5b. C6 and C7 rapidly bind to C5b, forming a C5b67 complex that possesses an unstable membrane-binding site; once bound to a membrane, this complex is stable. C8 and C9 then bind, forming the **membrane attack complex** (C5b6789) that creates a pore in the plasma membrane of the target cell (*see figure 31.14*). It is believed that the actual pore is a doughnut-shaped polymer of C9. (The perforin pores generated by cytotoxic T lymphocytes [figure 32.25] are somewhat similar to complement pores.) If the cell is eucaryotic, Na⁺ and H₂O enter through the pore leading to osmotic lysis of the cell. If the cell is a gram-negative bacterium, lysozyme from the blood enters through the pore and digests the peptidoglycan cell wall causing the bacterium to lyse osmotically. In contrast, gram-positive bacteria are resistant to the cytolytic action of the membrane attack complex because they lack an exposed outer membrane and the thick peptidoglycan prevents an attack on the plasma membrane. **The complement system and its functions** (pp. 714–18)

32.8 Acquired Immune Tolerance

Acquired immune tolerance refers to the body's ability to produce antibodies against nonself antigens such as microbial antigens, while “tolerating” (not producing antibodies against) self-antigens. Some of this tolerance arises early in embryonic life when immunologic competence is being established. Three general tolerance mechanisms have been proposed: negative selection by clonal deletion, the induction of anergy, and inhibition of the immune response by T-suppressor cells (p. 751).

Negative selection is one mechanism that produces immunologic tolerance. Negative selection by clonal deletion removes from the immune system T cells within the thymus that recognize any of the body's own antigens. T cell tolerance induced in the thymus and B cell tolerance in the bone marrow is called central tolerance. However, another mechanism is needed to prevent autoimmunity because many antigens are tissue-specific and are not present in the thymus or bone marrow.

Mechanisms occurring elsewhere in the body are collectively referred to as peripheral tolerance, which supplements central tolerance. Peripheral tolerance is thought to be based largely on incomplete activation signals given to the lymphocyte when it encounters self-antigens in the periphery of the body. This mechanism leads to a state of unresponsiveness called **anergy** [immunologists describe an inactive lymphocyte as “anergic” from the Greek *an*, negative, and *ergon*, without] that is associated with impaired intracellular signaling or apoptosis.

Many autoreactive B cells undergo clonal deletion or become anergic as they mature in the bone marrow. Negative selection occurs in the bone marrow if the B cells encounter large amounts of self-antigen, either in the soluble phase or as cell membrane constituents. The deletion of self-reactive B cells also takes place in secondary lymphoid tissue such as the spleen and lymph nodes. Since B cells recognize native antigen, there is no need for the MHC molecules in any of these processes. For those self-antigens that are present at relatively low concentrations, immunologic tolerance is often maintained only within the T-cell population. Nevertheless, this is sufficient to maintain tolerance because it denies the help essential for antibody production by self-reactive B cells.

1. Describe the three ways acquired immune tolerance develops in the vertebrate host.
2. How would you define the word anergy?

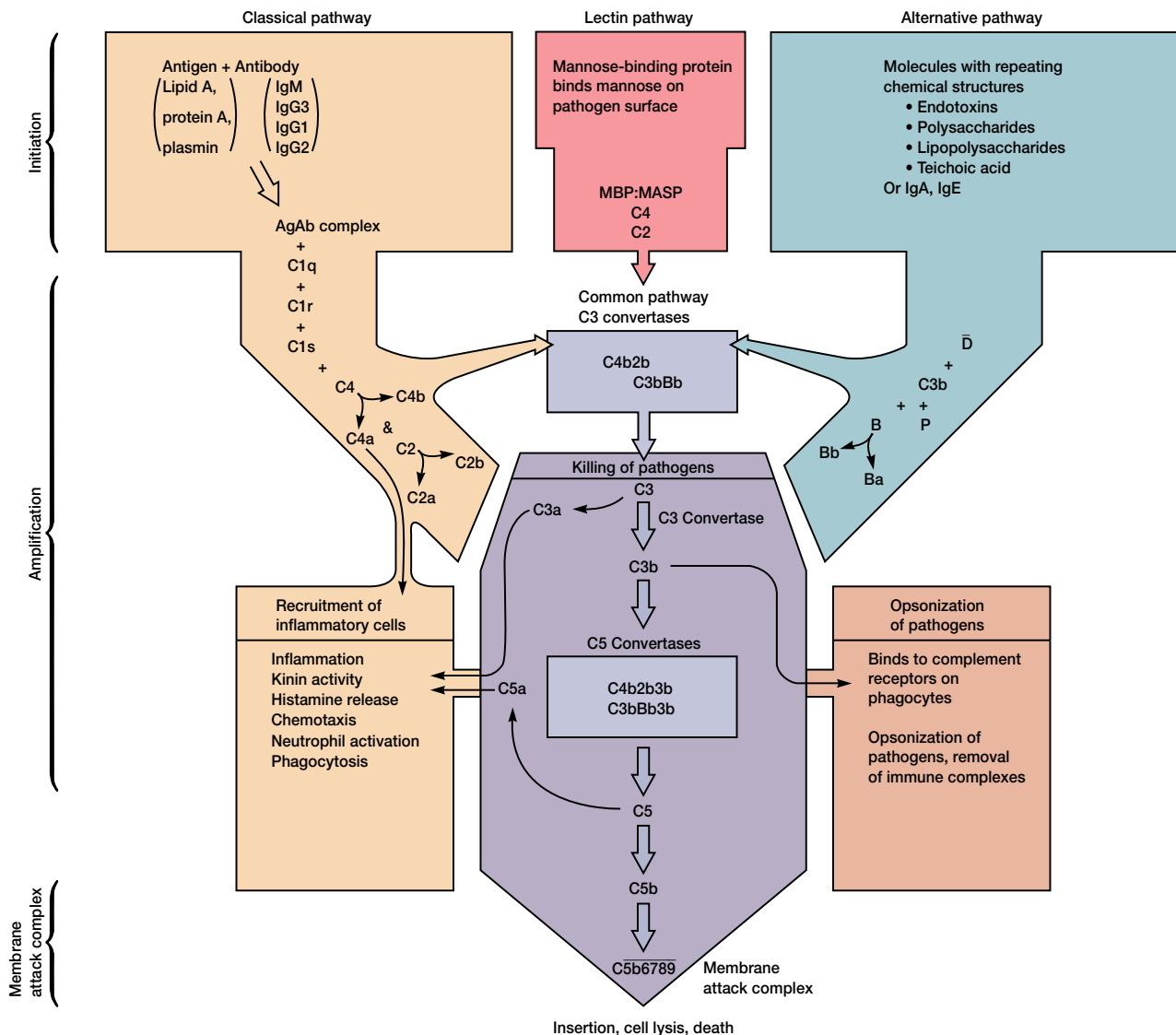


Figure 32.31 Complement Activation. Within each pathway the components are arranged in order of their activation and aligned opposite their functional and structural analogs in the opposite pathway. There are three pathways of complement activation: (1) the classical pathway (yellow), which is mediated by antibody; (2) the lectin-mediated pathway (red), which is triggered by a serum protein that binds to mannose residues of bacteria; and (3) the alternative pathway (blue), which is triggered directly on pathogen surfaces. All three generate a crucial enzymatic activity (common pathway of C3 convertases) that in turn generates the effector activity of complement. The three main consequences of complement activation are recruitment of inflammatory cells (left box), direct killing of pathogens by the membrane attack complex (middle box), and opsonization of pathogens (right box).

32.9 Summary: The Role of Antibodies and Lymphocytes in Resistance

The last part of chapter 31 presented nonspecific host immunity and this chapter described specific immunity. Although both the humoral and cellular arms of the specific immune response have been considered separately, it is important to understand that the response of a host to any particular pathogen may involve a com-

plex interaction between both the host and pathogen, as well as the components of both nonspecific and specific immunity to ensure a maximal survival advantage for the host. This last section summarizes the defense mechanisms vertebrate hosts have against viral and bacterial pathogens. At times, however, these defenses are not enough to protect the host. Chapter 34 continues this discussion by presenting those mechanisms that pathogens have evolved to circumvent many of the following defenses.

Immunity to Viral Infections

Resistance to virus infections involves humoral immunity, interferon sensitization of host cells, and cell-mediated immunity.

1. Antibodies can neutralize viruses by combining with them and interfering with their adsorption and entrance into cells (figure 32.29).
2. Antibodies enhance phagocytosis and destruction of viruses in much the same way as for bacteria (*see figure 31.15*).
3. Interferons are important in resistance when the target cell is reached immediately as in the case of colds and influenza. Interferon-stimulated cells shut down protein synthesis (*see figure 31.18*) and destroy viral mRNA. Some interferons also stimulate the activity of T cells (figure 32.24) and natural killer cells (*see figure 31.21*), thus accelerating the immune response to a viral infection.
4. Cell-mediated immunity to viruses is a major resistance mechanism when enveloped viruses modify host cell membranes and bud off from the surface (e.g., herpesvirus, poxvirus, influenza, mumps, measles, rabies, and rubella viruses). Activated lymphocytes can recognize and destroy virus-infected cells because the target cell's plasma membrane has been altered.
 - CTLs destroy virus-infected cells through FasL, and the production of granzymes and perforin that form channels through the plasma membrane of infected cells causing apoptosis and cytolysis (figure 32.25). The class I MHC proteins are involved in T-cell recognition of infected cells (figure 32.24). Cells displaying both viral antigens and the proper class I MHC will be destroyed. CTLs are also involved in the destruction of cancer cells (**immune surveillance**).
 - Natural killer cells (NK cells) are nonB, nonT lymphocytes that are active without any prior antigen exposure. Interferon and antibodies, however, will stimulate them to greater activity. They also are capable of destroying virus-infected and cancer cells (*see figure 31.21*). They are components of innate immunity and possess no antigen receptors.

Immunity to Bacterial Infections

Humoral immunity appears to be more important than cell-mediated immunity in the defense against most bacterial pathogens. Antibodies and complement attack pathogens in many ways.

1. IgG, and the C3b and C4b components of complement are opsonins. That is, they aid in the phagocytosis of bacteria by macrophages and granulocytes through the process called opsonization (*see figures 31.15 and 31.16*).
2. IgM and IgG will agglutinate bacterial pathogens, thus limiting their spread and enhancing the efficiency of phagocytosis (figure 32.30).
3. Antibodies can trigger complement attack on the wall of gram-negative bacteria by the classical pathway (figure 32.31). Once this pathway is activated, it forms a C5b-9 membrane attack complex (*see figure 31.14*) that forms a channel or pore in the bacterial cell, leading to lyses.
4. Complement C3a, C5a, and C5b67 also attract neutrophils and macrophages to a site of infection (*see figure 31.11*).
5. Antibodies called antitoxins (figure 32.29) bind to bacterial exotoxins and block their action (neutralization).
6. Cell-mediated immune responses by activated macrophages and T-cells (figure 32.1) are also important, particularly in resisting intracellular bacterial pathogens. Activated T cells secrete several cytokines, which have a variety of effects (*see figure 31.17 and table 31.4*).
 - The macrophage activating factor stimulates macrophages to become "angry" macrophages and more effectively phagocytose and destroy pathogens. Interferon- γ is a major macrophage activating factor.
 - The macrophage chemotactic factor and migration inhibition factor attract more macrophages and keep them in the area of infection after arrival.
 - Interleukin-2 (IL-2) stimulates the proliferation of activated T cells to increase the population of cells involved in the cell-mediated immune response. It also increases the effectiveness of cytotoxic T cells and NK cells by promoting the synthesis of immune interferon (γ -interferon) by T cells.

Summary

1. The specific immune response system consists of lymphocytes that can recognize foreign molecules (antigens) and respond to them. Two branches or arms of immunity are recognized: humoral (antibody-mediated) immunity and cellular (cell-mediated) immunity (figure 32.1).
2. Acquired immunity refers to the type of specific immunity that a host develops after exposure to a suitable antigen. It can be obtained actively or passively by natural or artificial means (figure 32.2).
3. An antigen is a substance that stimulates an immune response and reacts with the products of that response. Each antigen can have several antigenic determinant sites or epitopes that stimulate production of and combine with specific antibodies (figure 32.3). Haptens are small organic molecules that are not antigenic by themselves but can become antigenic if bonded to a larger carrier molecule (figure 32.4).
4. Antibodies or immunoglobulins are a group of glycoproteins present in the blood, tissue fluids, and mucous membranes of vertebrates. All immunoglobulins have a basic structure composed of four polypeptide chains (two light and two heavy) connected to each other by disulfide bonds (figure 32.7). In humans five immunoglobulin classes exist: IgG, IgA, IgM, IgD, and IgE (figures 32.10–32.14 and table 32.2).
5. Antibody diversity results from the rearrangement and splicing of the individual gene segments on the antibody coding chromosomes, somatic mutations, the generation of join diversity during splicing, and the independent assortment of light- and heavy-chain genes (figure 32.15 and table 32.3).
6. Immunologic specificity and memory is partly explained by the clonal selection theory (figure 32.18).

7. Specific antibodies can be produced naturally by immunized animals. The primary antibody response in a host occurs following initial exposure to the antigen. This response has lag, log, plateau, and decline phases. Upon secondary antigen challenge, the B cells mount a heightened and accelerated anamnestic response (**figure 32.19**).
8. Hybridomas result from the fusion of lymphoid cells with myeloma cells. These cells produce a single monoclonal antibody (**figure 32.20**). Monoclonal antibodies have many uses.
9. T cells are pivotal elements of the immune response. T cells have antigen-specific receptor proteins (**figure 32.21**).
10. Antigen-presenting cells—most of which are macrophages, dendritic cells, and B cells—take in foreign antigens or pathogens, process them, and present antigenic fragments complexed with MHC molecules (**figure 32.22**) to T-helper cells. These MHC molecules are proteins coded by a group of genes termed the major histocompatibility complex.
11. The cytotoxic T lymphocyte recognizes target cells such as virus-infected cells that have foreign antigens and class I MHC molecules on their surface. The CTLs then attack and destroy the target cells using the CD95 pathway and/or the perforin pathway (**figures 32.24 and 32.25**).
12. Regulator T cells control the development of effector T cells. Two types exist: T-helper cells (CD4⁺) and T-suppressor cells. There are three subsets of T-helper cells: T_H1, T_H2, and T_H0. T_H1 cells produce various cytokines and are involved in cellular immunity (**figure 32.24**). The T_H2 cells also produce various cytokines but are involved in humoral immunity. T_H0 cells are simply undifferentiated precursors of T_H1 and T_H2 cells.
13. B cells defend against antigens by differentiating into plasma cells that secrete antibodies into the blood and lymph, providing humoral or antibody-mediated immunity.
14. B cells can be stimulated to divide and/or differentiate to secrete antibody when triggered by the appropriate signals (**figures 32.24 and 32.27**).
15. B cells have receptor immunoglobulins on their plasma membrane surface that are specific for given antigenic determinants (**figure 32.28**). Contact with the antigenic determinant is required for the B cell to divide and differentiate into plasma cells and memory cells.
16. Various types of antigen-antibody reactions occur in vertebrates and initiate the participation of other body processes that determine the ultimate fate of the antigen. For example, the complement system can be activated, leading to cell lysis, phagocytosis, chemotaxis, or stimulation of the inflammatory response. Other defensive antigen-antibody interactions include toxin neutralization, viral neutralization, adherence inhibition, opsonization, and immune complex formation (**figures 32.29 and 32.30**).
17. Once activated, the classical complement pathway (**figure 32.31**) forms the membrane attack complex that can lyse antibody-coated cells.
18. Acquired immune tolerance is the ability of a host to react against nonself antigens while tolerating self-antigens. It can be induced in several ways.
19. Although both the humoral and cellular arms of the specific immune response have been considered separately, it is important to understand that the response of a vertebrate host to any particular pathogen may involve a complex set of responses. Both humoral and cellular immune responses can join nonspecific defenses to ensure a maximal survival advantage against viral and bacterial pathogens.

Key Terms

- | | | |
|---|--|---|
| acquired immune tolerance 758 | cytotoxic T (T _C) cell 748 | memory B cell 741 |
| acquired immunity 729 | cytotoxic T lymphocyte (CTL) 748 | monoclonal antibody (MAb) 743 |
| adjuvant 741 | domain 734 | myeloma cell 743 |
| agglutination reaction 756 | effector response 729 | naturally acquired active immunity 729 |
| agglutinin 756 | epitope 731 | naturally acquired passive immunity 729 |
| allotypes 734 | <i>fas</i> gene 750 | perforin pathway 750 |
| anamnestic response 743 | H-2 complex 745 | precipitation 756 |
| anergy 758 | hapten 731 | precipitin 756 |
| antibody or immunoglobulin (Ig) 734 | hemagglutination 756 | precipitin reaction 756 |
| antigen 731 | hemagglutinin 756 | regulator T cell 751 |
| antigen-binding fragment (Fab) 734 | human leukocyte antigen complex (HLA) 745 | secretory IgA (sIgA) 738 |
| antigen-presenting cell (APC) 745 | humoral (antibody-mediated) immunity 729 | serum 742 |
| antigenic determinant site 731 | hybridoma 743 | superantigen 732 |
| antisera 742 | idiotype 734 | T-cell antigen receptor (TCR) 745 |
| antitoxin 756 | IgA 736 | T-dependent antigen 753 |
| artificially acquired active immunity 730 | IgD 738 | T-helper (T _H) cell 751 |
| artificially acquired passive immunity 731 | IgE 738 | T _H 1 cell 751 |
| B-cell antigen receptor (BCR) 751 | IgG 736 | T _H 2 cell 751 |
| CD95 pathway 750 | IgM 736 | T _H 0 cell 751 |
| cellular (cell-mediated) immunity 729 | immune complex 756 | T-independent antigen 754 |
| classical complement pathway 758 | immune surveillance 760 | titer 742 |
| clonal selection theory 741 | immunotoxin 744 | toxin neutralization 756 |
| clone 741 | isotype 734 | T-suppressor (T _S) cell 751 |
| cluster of differentiation molecule (CD) 733 | J chain 736 | valence 731 |
| constant regions (C _L and C _H) 734 | major histocompatibility complex (MHC) 745 | variable regions (V _L and V _H) 734 |
| crystallizable fragment (Fc) 734 | membrane attack complex 758 | viral neutralization 756 |

Questions for Thought and Review

1. What is the difference between a nonspecific host defense mechanism and a specific host defense mechanism? Give examples.
2. What are the major differences between natural and artificial immunity and between active and passive immunity?
3. Compare humoral and cell-mediated immunity.
4. A person with AIDS has a low T-helper/T-cytotoxic cell ratio. What problem(s) does this create?
5. How do B and T cells cooperate in the immune response? What is the role of macrophages?
6. All antibodies are proteins. What protein property has favored their use as antibodies in preference to some other type of biomolecule?
7. How did the clonal selection theory inspire the development of monoclonal antibody techniques?
8. What is the difference in the kinetics of antibody formation in response to a first and second exposure to the same antigen?
9. What is the basis of immunologic memory?
10. Describe the derivation of B and T cells.
11. Discuss the role of the classical complement pathway in host defense.
12. What is the purpose of acquired immune tolerance?
13. Why are T-cell receptors an important link in T-cell functioning?

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Critical Thinking Questions

1. The placenta has many functions and one of them is as an immunosuppressive organ. Why is this necessary?
2. Why, evolutionarily, would sperm cells not express MHC on their surfaces?
3. What functionally important proteins are on the surface of B cells? T cells?
4. One of the main differences between an immature B cell and a plasma cell is the amount of endoplasmic reticulum. Explain which has more and why.
5. How would you distinguish a stem cell from a B cell at the DNA level?

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CHAPTER 33

Medical Immunology



This illustration is a wood engraving that first appeared in *Harper's Weekly* in 1885. During this year, Louis Pasteur (left) oversaw the administration of a rabies vaccine by a colleague (right) to Joseph Meister (center), a young boy who had been bitten repeatedly by a dog with rabies. As a result of the vaccination, Joseph Meister survived the bite and became a custodian in Louis Pasteur's laboratory—now called the Pasteur Institute.

Outline

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Concepts

1. Vaccines and immunizations are among the most cost-effective weapons for microbial disease prevention. Vaccines constitute one of the greatest achievements of modern medicine.
2. Some individuals experience harmful overreactions of the immune system known as hypersensitivities. There are four types: type I is characterized by the release of physiological mediators from IgE-bound mast cells and basophils; type II results from complement-dependent lysis of cells; type III involves the formation of immune complexes that are deposited on basement membranes; and type IV arises from the reaction of T_H1 cells, cytokines, and macrophages.
3. The union of antigen and antibody in vitro produces either a visible reaction or one that can be made visible in a variety of ways. These techniques can be used to identify viruses, microorganisms, and their products; to quantitate and identify antigens and antibodies; to follow the course of a disease; to determine the serotype of a microorganism; and to determine the amount of protection from disease an animal possesses.

Box 33.1

The First Immunizations

Since the time of the ancient Greeks, it has been recognized that people who have recovered from plague, smallpox, yellow fever, and various other infectious diseases rarely contract the diseases again. The first scientific attempts at artificial immunizations were made in the late eighteenth century by Edward Jenner (1749–1823): a country doctor from Berkley, Gloucestershire, England. Jenner investigated the basis for the widespread belief of the English peasants that anyone who had vaccinia (cowpox) never contracted smallpox. Smallpox was often fatal—10 to 40% of the victims died—and those who recovered had disfiguring pockmarks. Yet most English milkmaids, who were readily infected with cowpox, had clear skin because cowpox was a relatively mild infection that left no scars.

It was on May 14, 1796, that Jenner extracted the contents of a pustule from the arm of a cowpox-infected milkmaid, Sarah Nelmes, and injected it into the arm of eight-year-old James Phipps. As Jenner expected, immunization with the cowpox virus caused only mild symptoms in the boy. When he subsequently inoculated the boy with smallpox virus, the boy showed no symptoms of the disease.

Jenner then inoculated large numbers of his patients with cowpox pus, as did other physicians in England and on the European continent (see **Box figure**). By 1800 the practice known as vaccination had begun in America, and by 1805 Napoleon Bonaparte had ordered all French soldiers to be vaccinated.

Further work on immunization was carried out by Louis Pasteur (1822–1895). Pasteur discovered that if cultures of chicken cholera bacteria were allowed to age for two or three months the bacteria produced only a mild attack of cholera when inoculated into chickens. Somehow



Nineteenth-Century Physicians Performing Vaccinations on Children.

the old cultures had become less pathogenic (attenuated) for the chickens. He then found that fresh cultures of the bacteria failed to produce cholera in chickens that had been previously inoculated with old, attenuated cultures. To honor Jenner's work with cowpox, Pasteur gave the name vaccine to any preparation of a weakened pathogen that was used (as was Jenner's "vaccine virus") to immunize against infectious disease.

The three greatest discoveries of medicine in the last 200 years are antisepsis, antibiotics, and vaccines.

—Stanley A. Plotkin

Chapters 31 and 32 describe the various ways in which the immune system responds in vivo to protect the host by nonspecific resistance and specific immunity. Given the spectrum of diseases that result from inadequate or inappropriate immunity, clinical microbiologists are continually searching for ways to manipulate the immune response to benefit humankind. This chapter examines three areas of medical immunology in which this quest is being pursued: vaccines and immunizations, immune disorders, and antigen-antibody interactions in vitro.

33.1 Vaccines and Immunizations

Active immunization is the protection of susceptible humans and domestic animals from communicable diseases by the administration of vaccines (vaccination). A **vaccine** [Latin *vacca*, cow] is a

preparation from an infectious agent that is administered to humans and other animals to induce protective immunity. It may consist of a preparation of killed microorganisms; living, weakened (attenuated) microorganisms; inactivated bacterial toxins (toxoids); purified macromolecules; recombinant vectors (e.g., modified polio vaccine); or DNA vaccines (none approved yet) that are administered to an animal to induce immunity artificially.

The modern era of vaccines and vaccination began in 1798 with Edward Jenner's use of cowpox as a vaccine against smallpox (**Box 33.1**) and in 1881 with Louis Pasteur's anthrax vaccine. Vaccines for other diseases did not emerge until the latter part of the nineteenth century, when largely through a process of trial and error, methods for inactivating and attenuating microorganisms were developed and vaccines were produced. Vaccines were eventually developed against most of the epidemic diseases that had plagued western Europe and North America (diphtheria, measles, mumps, whooping cough, German measles, polio). Indeed, toward the end of the twentieth century it began to seem that the combination of vaccines and antibiotics would temper the problem of microbial infections. Such optimism was cut short by the emergence of new or previously unrecognized diseases (see section 37.8) and antibiotic resistance to old ones (see section 35.7). Nevertheless, vaccination is still one of the most cost-effective weapons for microbial disease prevention,

Table 33.1 Examples of Vaccines Used in the United States to Prevent Viral and Bacterial Diseases in Humans

Disease	Vaccine	Booster	Recommendation
Viral Diseases			
Chickenpox	Attenuated Oka strain (Varivax)	None	Children 12–18 months; older children who have not had chickenpox
Hepatitis A	Inactivated virus (Havrix)	6–12 months	International travelers
Hepatitis B	HB viral antigen (Engerix-B, Recombivax HB)	None	High-risk medical personnel; children, birth to 18 months and 11–12 years of age
Influenza	Inactivated virus or viral components	Yearly	Chronically ill individuals and those over 65
Measles, Mumps, Rubella	Attenuated viruses (combination MMR vaccine)	None	Children 15–19 months old
Poliomyelitis	Attenuated (oral poliomyelitis vaccine, OPV) or inactivated vaccine	Adults as needed	Children 2–3 years old
Rabies	Inactivated virus	None	For individuals in contact with wildlife, animal control personnel, veterinarians
Yellow fever	Attenuated virus	10 years	Military personnel and individuals traveling to endemic areas
Bacterial Diseases			
Anthrax	Extracellular components of unencapsulated <i>B. anthracis</i>	None	Agricultural workers, veterinary personnel
Cholera	Fraction of <i>Vibrio cholerae</i>	6 months	Individuals in endemic areas, travelers
Diphtheria, Pertussis, Tetanus	Diphtheria toxoid, acellular or killed <i>Bordetella pertussis</i> , tetanus toxoid (DPT vaccine)	10 years	Children 2–3 months old
<i>Haemophilus influenzae</i> type b	Polysaccharide-protein conjugate (HbCV) or bacterial polysaccharide (HbPV)	None	Children under 5 years of age
Lyme disease	Single surface antigen (lipoprotein Osp A)	None	Individuals in endemic area of microorganism or of vector
Meningococcal infections	Bacterial polysaccharides of serotypes A/C/Y/W-135	None	Military; high-risk individuals
Plague	Fraction of <i>Yersinia pestis</i>	Yearly	Individuals in contact with rodents in endemic areas
Pneumococcal pneumonia	Purified <i>S. pneumoniae</i> polysaccharide of 23 pneumococcal types	None	Adults over 50 with chronic disease
Tuberculosis	Attenuated <i>Mycobacterium bovis</i> (BCG vaccine)	3–4 years	Individuals exposed to TB for prolonged periods of time
Typhoid fever	Killed <i>Salmonella typhi</i>	3–4 years	Individuals in endemic areas
Typhus fever	Killed <i>Rickettsia prowazekii</i>	Yearly	Scientists and medical personnel in areas where typhus is endemic

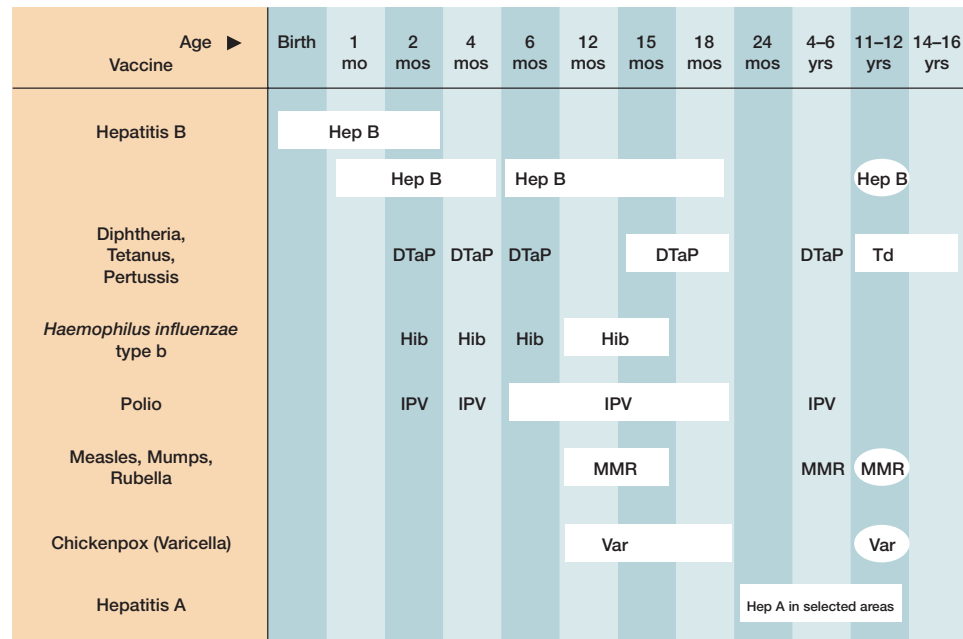
and vaccines arguably constitute one of the greatest achievements of modern medicine. **Table 33.1** summarizes the principal vaccines used to prevent viral and bacterial diseases in humans.

Vaccination of most children should begin at about 2 months of age. Before that age, they are protected by passive natural immunity. The recommended program of childhood immunizations for the United States is outlined in **figure 33.1**. Vaccination of adults depends on their risk group. For individuals living in close quarters (military personnel), the elderly, and those individuals with reduced immunity (transplant patients, people with sickle-cell anemia), vaccines for influenza, meningitis, and pneumonia are given. Depending on the country of travel, international travelers are routinely immunized against cholera, hepatitis, plague, polio, typhoid, typhus, and yellow fever. Health care personnel and people exposed through life-style patterns are immunized against hepatitis B. Immunization against anthrax is reserved for individuals (veterinarians or agricultural workers) who might come into contact with infected animals or products from the animals. Veterinarians, forest rangers, and others whose jobs involve contact with animals are vaccinated against rabies, plague,

and Lyme disease. Recently, suspected use of anthrax spores by terrorists or in biological warfare (*see section 37.10*) has prompted the military to vaccinate military personnel against anthrax. This procedure, however, is very controversial.

Passive immunization, artificially acquired passive immunity (*see figure 32.2*), can be produced by injecting an animal or human with preformed antibodies that have been produced in another animal, in another human, or in vitro (*see section 32.3*). This type of immunization is called passive because protection does not require participation of the recipient's immune system. Passive immunization is routinely administered to individuals exposed to certain microbial pathogens that cause diseases such as botulism, diphtheria, hepatitis, measles, rabies, and tetanus as well as to protect them against snake and spider bites (**table 33.2**). However, this form of immunization should be used only when absolutely necessary because of the risks involved such as developing anaphylaxis, serum sickness, or a type III hypersensitivity reaction. Furthermore, the protection lasts only as long as the antibody molecules survive in the recipient—months with antibodies from another human, but only weeks with antibodies from animals or in vitro methods.

Figure 33.1 Recommended Regimen and Indications for Vaccinations of Children in the United States. (Data from *The American Academy of Pediatrics*.)



Bars indicate range of recommended ages for immunization. Ovals indicate vaccine to be given if previous recommended doses were missed or given earlier than the recommended age.

Table 33.2 Immune Globulins and Antitoxins Used for Passive Immunization against Specific Organisms and Diseases in the United States

Organism/Disease	Agent ^a
Black widow spider bite	Horse antivenom
Snakebite	Horse antivenom
Acute respiratory failure (respiratory syncytial virus)	Monoclonal antibodies
Botulism	Botulinum antitoxin (equine)
Diphtheria	Diphtheria antitoxin (equine)
Hepatitis A and B	Pooled human immune gamma globulin
Measles	Pooled human immune gamma globulin
Rabies	Rabies immune globulin administered around the wound in addition to being injected intramuscularly
Tetanus	Tetanus immune globulin
Eczema	Vaccinia immune globulin
Immunocompromised individuals	Varicella-zoster immune globulin

^aImmune globulins and antitoxins are administered intramuscularly unless otherwise noted.

these diseases, the vaccine mimics the pathogen and causes an immune response similar to that raised by the pathogen. These vaccines have eradicated smallpox, pushed polio to the brink of extinction, and spared countless individuals from hepatitis A and B, measles, rotavirus disease, tetanus, typhus, and other dangerous diseases.

In contrast, successful vaccines have yet to be developed for too many deadly or debilitating diseases (AIDS, herpes, hepatitis C, malaria, tuberculosis) because they are due to chronic infections. In a chronic infection, the pathogen is able to evade the immune system or subvert it into making ineffective responses. Successful vaccines against these chronic diseases must be able to stimulate immune responses that are similar to those resulting from most natural exposures to the pathogen. To date, this has been an unsolvable challenge in medical immunology and vaccine development.

In the remainder of this section, various approaches to the design of vaccines, both currently used vaccines and experimental ones, are described and examined in terms of their ability to induce humoral and cell-mediated immunity and the production of memory cells. **Vaccinomics**, the application of genomics and bioinformatics to vaccine development, is bringing a fresh approach to the Herculean problem of making vaccines against various microorganisms and parasites.

Types of Vaccines and Their Characteristics

As previously noted, the diseases for which we already have vaccines are ones in which the infection is acute and either resolves in several weeks or causes death of the infected individual. In the absence of vaccination, many individuals would survive as a result of the immune system defeating the invader. For

Whole-Organism Vaccines

Many of the current vaccines in use for humans that are effective against viral and bacterial diseases consist of whole microorganisms that are either inactivated (killed) or **attenuated** (live but avirulent) (table 33.1). These are termed **whole-organism vaccines**. The major characteristics of these vaccines are compared

Table 33.3 A Comparison of Inactivated (Killed) and Attenuated (Live) Vaccines

Major Characteristic	Inactivated Vaccine	Attenuated Vaccine
Booster shots	Multiple boosters required	Only a single booster
Production	Virulent microorganism inactivated by chemicals or irradiation	Virulent microorganism grown under adverse conditions or passed through different hosts until avirulent
Reversion tendency	None	May revert to a virulent form
Stability	Very stable, even where refrigeration is unavailable	Less stable
Type of immunity induced	Humoral	Humoral and cell-mediated

Source: Adapted from Goldsby, T. J. Kindt, and B. A. Osborne, *Kuby Immunology*. 2000, New York: W. H. Freeman.

and contrasted in **table 33.3**. Inactivated vaccines are effective, but they often require several boosters and normally do not adequately stimulate cell-mediated immunity or secretory IgA production. In contrast, attenuated vaccines usually are given in a single dose and stimulate both humoral and cell-mediated immunity.

Even though whole-organism vaccines are considered the “gold standard” of existing vaccines, they can be problematic in their own way. For example, whole-organism vaccines fail to shield against some diseases. Attenuated vaccines that do work can also cause full-blown illness in individuals whose immune system is compromised (AIDS patients, cancer patients undergoing chemotherapy, the elderly). These same individuals may also contract the disease from healthy people who have been vaccinated recently. Moreover, attenuated viruses can at times mutate in ways that restore virulence, as has happened in some monkeys given an attenuated simian form of the AIDS virus. In the case of very lethal diseases, the risk of reversion to virulence is intolerable. Whole-organism vaccines, whether live or dead, have another big drawback. Since they are composed of complete pathogens, they retain molecules that are not involved in evoking immunity. These molecules, as well as contaminants that are unavoidable by-products of the manufacturing process, can trigger allergic or other disruptive reactions.

Purified Macromolecules as Vaccines

A few of the common risks associated with whole-organism vaccines can be avoided by using only specific, purified macromolecules derived from pathogenic microorganisms. Currently, there are three general forms of **macromolecule vaccines**: (1) capsular polysaccharides, (2) recombinant surface antigens, and (3) inactivated exotoxins called **toxoids** (see p. 796). The purified macromolecule vaccines that are currently available for humans are listed in **table 33.4**.

Recombinant-Vector Vaccines

Genetic vaccines are quite different in structure from whole-organism vaccines. It is now possible to isolate genes that encode major antigens from a pathogen and insert them into nonvirulent viruses or bacteria. The vaccines are usually delivered by needle injection or by a device called a gene gun. The attenuated microorganism serves as a vector, replicating within the host and ex-

Table 33.4 Purified Macromolecule Vaccines Currently Available for Human Use

Type of Purified Macromolecule (Disease or Microorganism)	Form of Vaccine
Capsular polysaccharide <i>Haemophilus influenzae</i> type b	Polysaccharide-protein conjugate (HbCV) or bacterial polysaccharide (HbPV)
<i>Neisseria meningitidis</i>	Polysaccharides of serotypes A/C/Y/W-135
<i>Streptococcus pneumoniae</i>	23 distinct capsular polysaccharides
Surface antigen Hepatitis B	Recombinant surface antigen (HbsAg)
Toxoids Diphtheria Tetanus	Inactivated exotoxin Inactivated exotoxin

pressing the gene product of the pathogen-encoded antigenic proteins. The antigens can elicit humoral immunity when they escape from the vector, and they can also elicit cellular immunity when they are broken down and properly displayed on the cell surface (just as occurs when cells harbor an active pathogen).

Recently several microorganisms have been used in the production of these **recombinant-vector vaccines**. Examples include adenovirus, vaccinia virus, canarypox virus, attenuated poliovirus, and attenuated strains of *Salmonella* and *Mycobacterium*.

DNA Vaccines

A more complicated genetic vaccine to emerge in recent years is the DNA vaccine. A **DNA vaccine** elicits protective immunity against a microbial pathogen by activating both branches of the immune system: humoral and cellular. Long-lasting memory cells also are generated.

The immunization procedure begins with the injection into muscle of a plasmid preparation that contains genes for pathogen antigens. The plasmids are taken up by muscle cells, enter the cell nuclei, and express their antigen genes. The muscle cells commence protein synthesis and produce the pathogen's antigenic proteins.

In the humoral response, the antigenic proteins are released from the muscle cell and bind to B-cell antigen receptors. At the same time, antigen-presenting cells ingest the antigenic proteins, break them down, and display the fragments on a class II MHC molecule. Helper T cells then recognize the antigen and secrete T_H2 cytokines that activate B cells containing bound antigen. The activated B cells multiply and differentiate into plasma cells that release antibodies which bind to the pathogen and neutralize it or mark it for destruction. Other B cells become memory cells that will protect the host against future infections with the same pathogen.

The cellular immune response begins when the muscle cells display the same antigenic proteins or protein fragments (peptides) on MHC class I molecules. The plasmid vaccine vector also becomes incorporated into antigen-presenting cells, which synthesize the encoded antigens and express fragments of the antigens on MHC I molecules along with co-stimulatory molecules. Cytotoxic T cells recognize these signals and are additionally stimulated by cytokines (this time, T_H1 cytokines) from helper T cells. The activated cytotoxic T cells multiply and attack cells infected by the pathogen. Some activated T cells also develop into memory T cells and protect against future infections.

At present, there are human trials under way with several different DNA vaccines against malaria, AIDS, influenza, hepatitis B, and herpesvirus. Vaccines against a number of cancers (lymphomas, prostate, colon) are also being tested.

1. What is a vaccine? Briefly describe and contrast active and passive immunization.
2. Describe the major differences between inactivated and attenuated vaccines. Discuss the advantages and disadvantages of using attenuated microorganisms as vaccines.
3. Give the three types of purified macromolecules that are currently used as vaccines.
4. What are some advantages of genetically engineered vaccines? Some disadvantages?

33.2 Immune Disorders

Chapters 31 and 32 presented the awesome power of the immune system in protecting the host against the continual onslaught of microbial invaders and cancer cells. Like any system in a vertebrate host, disorders (malfunctions) also occur in the immune system. Immune disorders can be categorized as hypersensitivities, autoimmune diseases, transplantation (tissue) rejection, and immunodeficiencies. Each of these immune disorders is now discussed.

Hypersensitivities

Hypersensitivity is an exaggerated immune response that results in tissue damage and is manifested in the individual on second or subsequent contact with an antigen. Hypersensitivity reactions can be classified as either immediate or delayed. Obviously immediate reactions appear faster than delayed ones, but the main difference between them is in the nature of the immune response

to the antigen. Realizing this fact, Peter Gell and Robert Coombs developed a classification system for reactions responsible for hypersensitivities in 1963. Their system correlates clinical symptoms with information about immunologic events that occur during hypersensitivity reactions. The Gell-Coombs classification system divides hypersensitivity into four types: I, II, III, and IV.

Type I Hypersensitivity

Allergic reactions occur when an individual who has produced IgE antibody in response to an innocuous antigen (**allergen**) subsequently encounters the same allergen. An **allergy** [Greek *allos*, other and *ergon*, work] is one kind of type I hypersensitivity reaction.

Type I hypersensitivity is characterized by an allergic reaction occurring immediately following an individual's second contact with the responsible antigen (the allergen). Upon initial exposure to a soluble allergen, B cells are stimulated to differentiate into plasma cells and produce specific IgE with the help of T cells (**figure 33.2**). This IgE is sometimes called a **reagin**, and the individual has a hereditary predisposition for its production. Once synthesized, IgE binds to the Fc receptors of mast cells (basophils and eosinophils can also be activated) and sensitizes these cells, making the individual allergic to the allergen. When a second exposure to the allergen occurs, the allergen attaches to the surface-bound IgE on the sensitized mast cells causing degranulation. Degranulation releases physiological mediators such as histamine, leukotrienes, heparin, prostaglandins, PAF (platelet-activation factor), ECF-A (eosinophil chemotactic factor of anaphylaxis), and proteolytic enzymes. These mediators trigger smooth muscle contractions, vasodilation, increased vascular permeability, and mucous secretion. The inclusive term for these responses is **anaphylaxis** [Greek *ana*, up, back, again; and *phylaxis*, protection]. Anaphylaxis can be divided into systemic and localized reactions.

Systemic anaphylaxis is a generalized response that occurs when an individual sensitized to an allergen receives a subsequent exposure to it. The reaction is immediate due to the large amount of mast cell mediators released over a short period. Usually there is respiratory impairment caused by smooth muscle constriction in the bronchioles. The arterioles dilate, which greatly reduces arterial blood pressure and increases capillary permeability with rapid loss of fluid into the tissue spaces. Because of these reactions the individual can die within a few minutes from reduced venous return, asphyxiation, reduced blood pressure, and circulatory shock. Common examples of allergens that can produce systemic anaphylaxis include drugs (penicillin), passively administered antisera, peanuts, and insect venom from the stings or bites of wasps, hornets, or bees (**figure 33.3**).

Localized anaphylaxis is called an atopic ("out of place") allergy. The symptoms that develop depend primarily on the route by which the allergen enters the body. **Hay fever** (allergic rhinitis) is a good example of an atopic allergy involving the upper respiratory tract. Initial exposure involves airborne allergens—such as plant pollen, fungal spores, animal dander, and house dust mites—that sensitize mast cells located within the mucous membranes. Reexposure to the allergen causes the typical localized anaphylactic response: itchy and tearing eyes, congested nasal

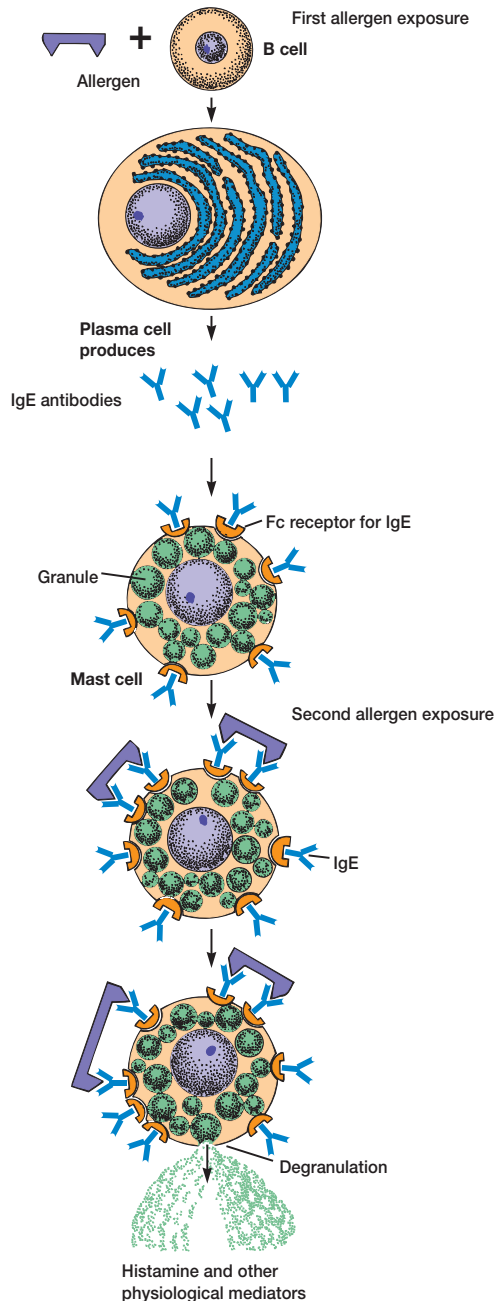


Figure 33.2 Type I Hypersensitivity. This type of hypersensitivity occurs when IgE antibodies attach to mast cells. The combination of these antibodies with allergens stimulates the mast cell (or basophil) to degranulate and produce the physiological mediators that cause the anaphylactic reaction, asthma, or hay fever.

passages, coughing, and sneezing. Antihistamine drugs are used to help alleviate these symptoms.

Bronchial asthma (asthma means panting) is an example of an atopic allergy involving the lower respiratory tract. Common

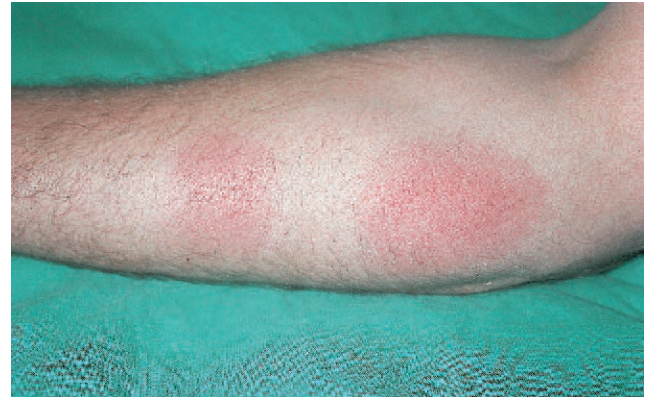


Figure 33.3 The Anaphylactic Response to Bee Venom. This person has been stung on the arm by a bee, leading to type I hypersensitivity in a generalized area.

allergens are the same as for hay fever. In bronchial asthma, however, the air sacs (alveoli) become overdistended and fill with fluid and mucus; the smooth muscle contracts and narrows the walls of the bronchi. Bronchial constriction produces a wheezing or whistling sound during exhalation. Symptomatic relief is obtained from bronchodilators that help relax the bronchial muscles, and from expectorants and liquefacients that dissolve and expel mucous plugs that accumulate.

Allergens that enter the body through the digestive system may cause food allergies. **Hives** (eruptions of the skin) are a good diagnostic sign of a true food allergy. Once established, type I food allergies are usually permanent but can be partially controlled with antihistamines or by avoidance of the allergen.

Skin testing can be used to identify the allergen responsible for allergies. These tests involve inoculating small amounts of suspect allergen(s) into the skin. Sensitivity to the antigen is shown by a rapid inflammatory reaction characterized by redness, swelling, and itching at the site of inoculation (**figure 33.4**). The affected area in which the allergen-mast cell reaction takes place is called a wheal and flare reaction site.

Once the responsible allergen has been identified, the individual should avoid contact with it. At times this is not possible, and **desensitization** is warranted. This procedure consists of a series of allergen doses injected beneath the skin to stimulate the production of IgG antibodies rather than IgE antibodies. The circulating IgG antibodies can then act as blocking antibodies to intercept and neutralize allergens before they have time to react with mast cell-bound IgE. Recent evidence suggests that suppressor T-cell activity also may cause a decrease in IgE synthesis. Desensitizations are about 65 to 75% effective in individuals whose allergies are caused by inhaled allergens.

Type II Hypersensitivity

Type II hypersensitivity is generally called a cytolytic or cytotoxic reaction because it results in the destruction of host cells,



(a)



(b)

Figure 33.4 In Vivo Skin Testing. (a) Skin prick tests with grass pollen in a person with summer hay fever. Notice the various reactions with increasing dosages (from top to bottom). (b) Skin patch test. The surface of the skin (left) is abraded and the suspect allergic extract placed on the skin. After 48 hours (center) it is eczematous and positive for the suspect antigen.

either by lysis or toxic mediators. In type II hypersensitivity, IgG or IgM antibodies are directed against cell surface or tissue-associated antigens. They usually stimulate the complement pathway and a variety of effector cells (**figure 33.5**). The antibodies interact with complement (C1q) and the effector cells through their Fc regions. The damage mechanisms are a reflection of the normal physiological processes involved in interaction of the immune system with pathogens. A classic example of type II hypersensitivity is that resulting when a person receives a transfusion with blood from a donor with a different blood group.

Type III Hypersensitivity

Type III hypersensitivity involves the formation of immune complexes (**figure 33.6a**). Normally these complexes are removed effectively by the fixed monocytes and macrophages of

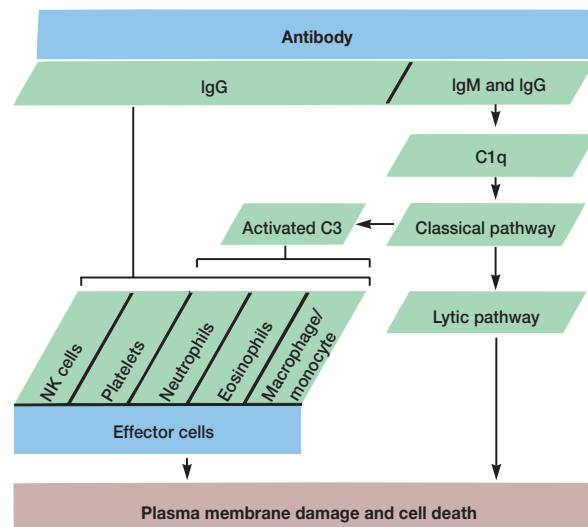


Figure 33.5 Type II Hypersensitivity. The action of antibody occurs through effector cells or the membrane attack complex, which damages target cell plasma membranes, causing cell destruction.

the monocyte-macrophage system. In the presence of excess amounts of some soluble antigens, the antigen-antibody complexes may not be efficiently removed. Their accumulation can lead to a hypersensitivity reaction from complement that triggers a variety of inflammatory processes. This inflammation causes damage, especially of blood vessels (vasculitis; **figure 33.6b**), kidney glomerular basement membranes (glomerulonephritis), joints (arthritis), and skin.

Diseases resulting from type III reactions can be placed into three groups. First, a persistent viral, bacterial, or protozoan infection, together with a weak antibody response, leads to chronic immune complex formation and eventual deposition of the complex in host tissues. Second, the continued production of autoantibody to self-antigen during an autoimmune disease can lead to prolonged immune complex formation. This overloads the monocyte-macrophage system, and tissue deposition of the complexes occurs (e.g., in the disease **systemic lupus erythematosus**). Third, immune complexes can form at body surfaces (such as the lungs), following repeated inhalation of allergens from molds, plants, or animals. For example, in Farmer's lung disease, an individual has circulating antibodies to fungi after being exposed repeatedly to moldy hay. These antibodies are primarily IgG. When the allergens (fungal spores) enter the alveoli of the lungs, local immune complexes form, leading to inflammation.

Some group A streptococcal infections can produce an immunologically mediated acute glomerulonephritis. Although the mechanism is not completely understood, it is believed that complexes of antibody and streptococcal antigen are deposited within the kidney glomeruli and generate a type III hypersensitivity reaction.

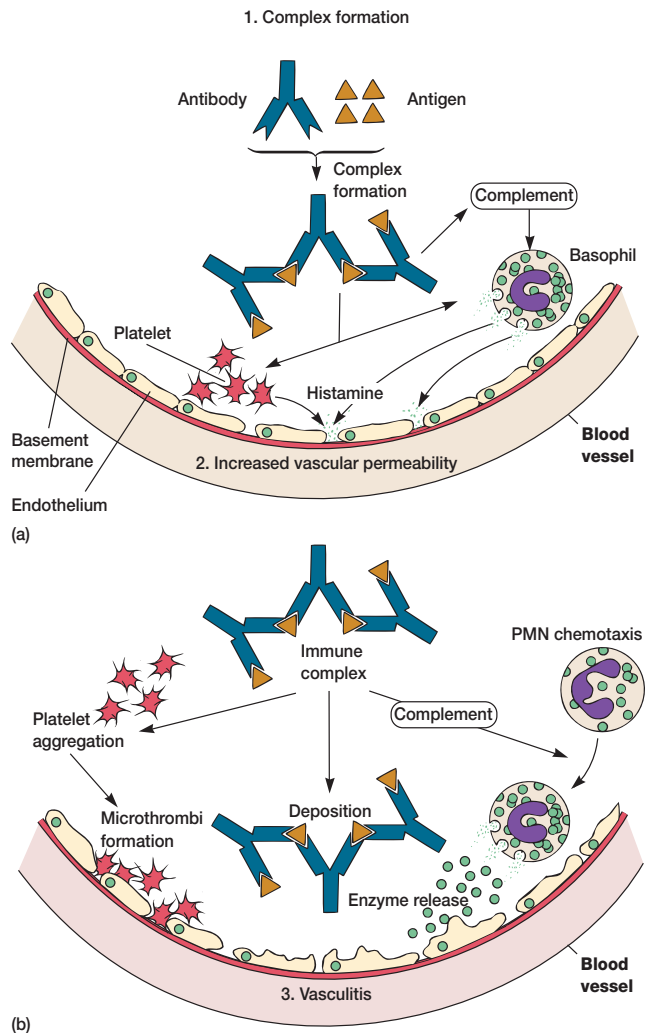


Figure 33.6 Type III Hypersensitivity. Deposition of immune complexes in blood vessel walls. (a) Antibody and antigen combine to form immune complexes. These activate complement, which causes basophils and platelets to degranulate and release histamine and other mediators. These mediators increase vascular permeability. (b) The increased permeability allows the immune complexes to be deposited in the blood vessel wall. This induces platelet aggregation to form microthrombi (blood clots) on the vessel wall. PMNs, stimulated by complement, degranulate causing enzymatic damage to the blood vessel wall.

Type IV Hypersensitivity

Type IV hypersensitivity involves delayed T-cell-mediated immune reactions. A major factor in the type IV reaction is the time required for a special subset of T_H1 cells (often called delayed-type hypersensitivity [T_{DTH}] cells) to migrate to and accumulate near the antigens. This usually takes a day or more.

Type IV reactions occur when antigens, especially those binding to tissue cells, are phagocytosed by macrophages and then pre-

sented to receptors on the T_H1 cell surface in the context of class I MHC. Contact between the antigen and T_H1 cell causes the cell to proliferate and release cytokines. Cytokines attract lymphocytes, macrophages, and basophils to the affected tissue. Extensive tissue damage may result. Examples of type IV hypersensitivities include tuberculin hypersensitivity (the **TB skin test**), allergic contact dermatitis, some autoimmune diseases, transplantation rejection, and killing of cancer cells.

In tuberculin hypersensitivity a partially purified protein called tuberculin, which is obtained from the bacillus that causes tuberculosis (*Mycobacterium tuberculosis*; see section 39.1), is injected into the skin of the forearm (**figure 33.7a**). The response in a tuberculin-positive individual begins in about 8 hours, and a redened area surrounding the injection site becomes indurated (firm and hard) within 12 to 24 hours. The T_H1 cells that migrated into the injection site are responsible for the induration. The reaction reaches its peak in 48 hours and then subsides. The size of the induration is directly related to the amount of antigen that was introduced and to the degree of hypersensitivity of the tested individual. Other microbial products used in type IV skin testing are histoplasmin for histoplasmosis, coccidioidin for coccidioidomycosis, lepromin for leprosy, and brucellergen for brucellosis.

Allergic contact dermatitis is caused by haptens (see figure 32.4) that combine with proteins in the skin to form the allergen that elicits the immune response. The haptens are the antigenic determinants, and the skin proteins are the carrier molecules for the haptens. Examples of these haptens include cosmetics, plant materials (catechol molecules from poison ivy and poison oak; figure 33.7b), topical chemotherapeutic agents, metals, and jewelry (especially jewelry containing nickel).

Several important chronic diseases involve cell and tissue destruction by type IV hypersensitivity reactions. These diseases are caused by viruses, mycobacteria, protozoa, and fungi that produce chronic infections in which the macrophages and T cells are continually stimulated. Examples are leprosy, tuberculosis, leishmaniasis, candidiasis, and herpes simplex lesions.

1. Discuss the mechanism of type I hypersensitivity reactions and how these can lead to systemic and localized anaphylaxis.
2. What causes a wheal and flare reaction site?
3. Why are type II hypersensitivity reactions called cytolytic or cytotoxic?
4. What characterizes a type III hypersensitivity reaction? Give an example.
5. Characterize a type IV hypersensitivity reaction.
6. What is the TB skin test used for?

Autoimmune Diseases

As discussed earlier, the body is normally able to distinguish its own self-antigens from foreign nonself antigens and does not mount an immunologic attack against the former. This phenomenon is called immune tolerance. At times the body loses tolerance and mounts an abnormal immune attack, either with antibodies or T cells, against a person's own self-tissue antigens.

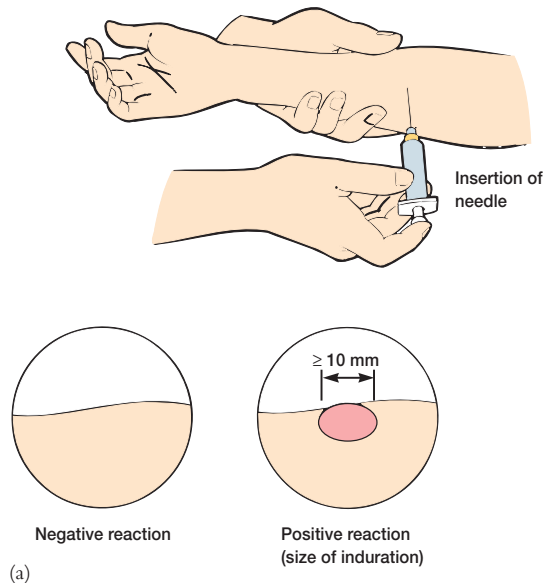
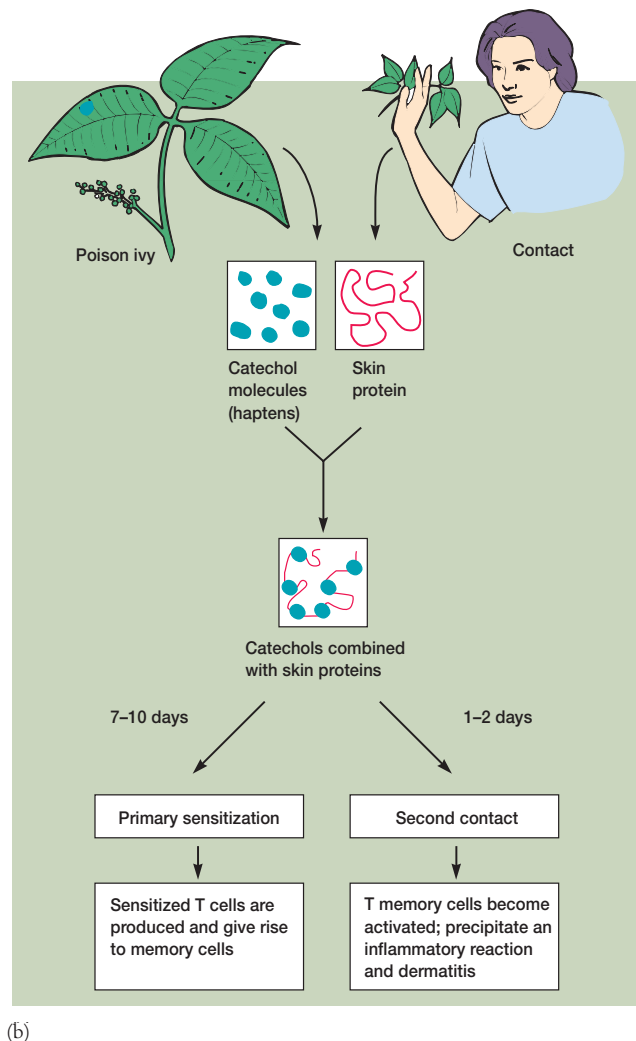


Figure 33.7 Mechanism of Type IV Hypersensitivity. (a) The tuberculin skin test. A positive reaction is one in which the diameter of induration is 10 mm or more. (b) In this example of contact dermatitis to poison ivy, a person initially becomes exposed to the catechol molecules from the poison ivy plant. The catechols combine with high molecular weight skin proteins and act as haptens. After 7–10 days sensitized T cells are produced and give rise to memory T cells. Upon second contact, the catechols bind to the same skin proteins, and the T memory cells become activated in only 1–2 days, leading to the inflammatory reaction (contact dermatitis).



It is important to distinguish between autoimmunity and autoimmune disease. Autoimmunity often is benign, whereas autoimmune disease often is fatal. **Autoimmunity** is characterized only by the presence of serum autoantibodies. It is a normal consequence of aging; is readily inducible by infectious agents, organisms, or drugs; and is potentially reversible (it disappears when the offending “agent” is removed or eradicated). **Autoimmune disease** results from the activation of self-reactive T and B cells that, following stimulation by genetic or environmental triggers, cause actual tissue damage (table 33.5).

Four factors influence the development of autoimmune disease. Two major factors are genetic and viral. The third factor is endocrine; the hormone estrogen promotes autoimmune disease, and androgens act as natural immunosuppressive agents. These physiological modulatory effects of sex hormones acting on normal immune responses partially explain the marked female predominance of autoimmune diseases. The fourth factor is psy-

choneuroimmunological—that is, the influence of stress and neurochemicals on the immune response. Overall, all four of these factors can affect gene expression, which directly or indirectly interferes with important immunoregulatory actions.

Although their causal mechanism is not well known, these diseases are more common in older people and may involve viral or bacterial infections. Some investigators believe that the release of abnormally large quantities of antigens may occur when the infectious agent causes tissue damage. The same agents also may cause body proteins to change into forms that stimulate antibody production or T-cell activation. Simultaneously the activity of T-suppressor cells, which normally limits this type of reaction, seems to be repressed. Many autoimmune diseases have a genetic component. For example, there is a well-known association between an individual’s susceptibility to Graves’ disease and multiple sclerosis and specific determinants on the major histocompatibility complex.

Table 33.5 Some Autoimmune Diseases in Humans

Disease	Autoantigen	Pathophysiology
Type II Hypersensitivity: Antibody to Cell-Surface or Cell Matrix Antigens (Cytotoxic)		
Acute rheumatic fever	Streptococcal cell wall antigens; antibodies cross-react with cardiomyocytes	Arthritis, scarring of heart valves, myocarditis
Autoimmune hemolytic anemia	Rh blood group, I antigen	Red blood cells are destroyed by complement and phagocytosis, anemia
Autoimmune thrombocytopenia purpura	Platelet integrin	Perfuse bleeding
Goodpasture's syndrome	Basement membrane collagen	Glomerulonephritis, pulmonary hemorrhage
Grave's disease	Thyroid-stimulating hormone receptor	Hyperthyroidism
Myasthenia gravis	Acetylcholine receptor	Progressive muscular weakness
Pemphigus vulgaris	Cadherin in epidermis	Skin blisters
Type III Hypersensitivity: Immune Complex Disease		
Systemic lupus erythematosus	DNA, histones, ribosomes	Arthritis, glomerulonephritis, vasculitis
Type IV Hypersensitivity: T-Cell-Mediated Disease		
Type 1 diabetes mellitus	Pancreatic beta cell antigen	Beta cell destruction
Multiple sclerosis	Myelin basic protein	Demyelination of axons
Rheumatoid arthritis ^a	Unknown synovial joint antigen	Joint inflammation and destruction

^a Type III hypersensitivity processes also may be involved.

Transplantation (Tissue) Rejection

Tissue transplant rejection is the third area (after hypersensitivity and autoimmunity) in which the immune system can act detrimentally. It is occasionally desirable to replace a nonfunctional or damaged body part by transplanting a tissue or organ from one person to another. Transplants between genetically different individuals within a species are termed **allografts** [Greek *allos*, other]. Some transplants do not stimulate an immune response. For example, a transplanted cornea is rarely rejected since lymphocytes do not circulate into the anterior chamber of the eye. This site is considered an immunologically privileged site. Another example of a privileged tissue transplant is the heart valve transplanted from a pig to a human. Such a graft between different species is termed a **xenograft** [Greek *xenos*, strayed].

With allografts, there is the possibility that the recipient's cells will recognize the donor's tissues as foreign. This triggers the recipient's immune mechanisms, which may destroy the donor tissue. Such a response is called a tissue rejection reaction. A tissue rejection reaction can occur by two different mechanisms. First, foreign major histocompatibility complex (MHC) class II molecules on the graft stimulate host T-helper cells to aid cytotoxic T cells in graft destruction (**figure 33.8a**). Cytotoxic T cells recognize the graft through the foreign MHC class I molecules. A second mechanism involves the T-helper cells reacting to the graft and releasing cytokines (**figure 33.8b**). The cytokines stimulate macrophages to enter the graft and destroy it.

As presented in **figure 33.8**, the major histocompatibility complex molecules play a dominant role in tissue rejection reactions because of their unique association with the recognition system of T cells. Unlike antibodies, T cells cannot recognize or react directly with non-MHC molecules (viruses, allergens). They recognize these molecules only in association with, or complexed to, an MHC molecule. There are two classes of MHC molecules. Class I MHC molecules are present on every cell in the body and consequently are important targets of the rejection reaction. Class II MHC molecules are involved in T-helper cell activation as previously discussed. The greater the antigenic difference between class I molecules of the recipient and donor tissues, the more rapid and severe the rejection reaction is likely to be. However, the reaction can sometimes be minimized if recipient and donor tissues are matched as closely as possible.

Immunosuppression in organ transplant cases also can lead to **graft-versus-host disease**. This occurs when the transplanted tissue contains immunocompetent cells that recognize host antigens and attack the host. The immunosuppressed recipient cannot control the response of the grafted tissue. Graft-versus-host disease is a common problem in allogeneic bone marrow transplants. The transplanted bone marrow contains many mature post-thymic T cells. These cells recognize the host MHC antigens and attack the immunosuppressed recipient's normal tissue cells. Currently one way to prevent graft-versus-host disease is to deplete the bone marrow of mature T cells by using immunosuppressive techniques. Examples include drugs that

Figure 33.8 Graft Cell

Destruction. (a) Foreign MHC class II molecules on the graft cell stimulate host T-helper cells to help cytotoxic T cells destroy the target graft cell. Cytotoxic T cells recognize the graft cell by its foreign MHC class I molecules. (b) T-helper cells reacting to the graft cell release cytokines that stimulate macrophages to enter the graft and destroy it by cytotoxic action.

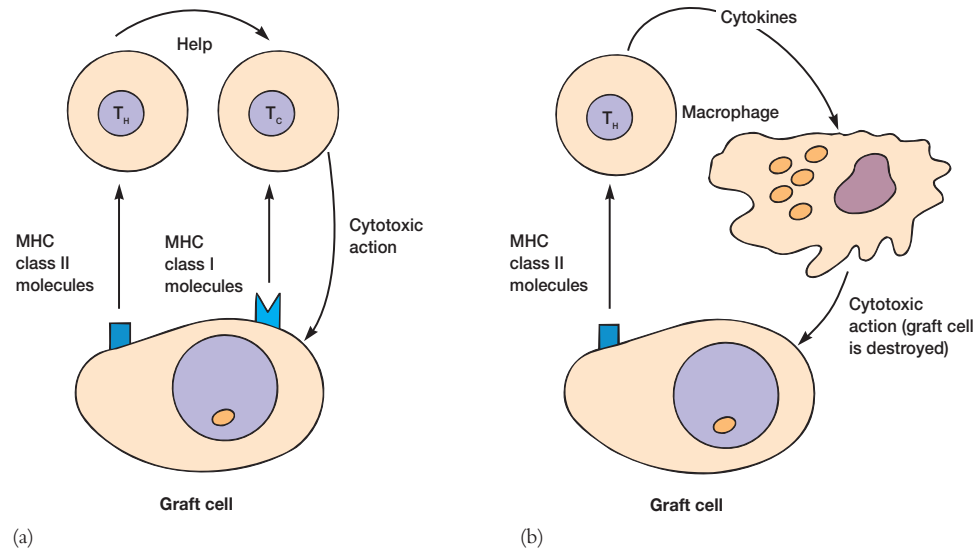


Table 33.6 Some Congenital Immune Deficiencies in Humans

Condition	Symptoms	Cause
Chronic granulomatous disease	Defective monocytes and neutrophils leading to recurrent bacterial and fungal infections	Failure to produce reactive oxygen intermediates due to defective NADPH oxidase
X-linked agammaglobulinemia	Plasma cell or B-cell deficiency and inability to produce adequate specific humoral antibodies	Defective B cell differentiation due to loss of tyrosine kinase
DiGeorge syndrome	T-cell deficiency and very poor cell-mediated immunity	Lack of thymus or a poorly developed thymus
Severe combined immunodeficiency disease (SCID)	Both antibody production and cell-mediated immunity impaired due to a great reduction of B- and T-cell levels	Various mechanisms (e.g., defective B- and T-cell maturation because of X-linked gene mutation; absence of adenosine deaminase in lymphocytes)

attack T cells (azathioprine and cyclophosphamide), immunosuppressive drugs (cyclosporin), anti-inflammatory drugs (corticosteroids), irradiation of the lymphoid tissue, and antibodies directed against T-cell antigens.

Immunodeficiencies

Defects in one or more components of the immune system can result in its failing to recognize and respond properly to antigens. Such **immunodeficiencies** can make a person more prone to infection than those people capable of a complete and active immune response. Despite the increase in knowledge of functional derangements and cellular abnormalities in the various immunodeficiency disorders, the fundamental biological errors responsible for them remain largely unknown. To date, most genetic errors associated with these immunodeficiencies are located on the X chromosome and produce primary or congenital immunodeficiencies (**table 33.6**). Other immunodeficiencies can be acquired because of infections by immunosuppressive microorganisms (chronic mucocutaneous candidiasis) or by some viruses (HIV).

1. What is an autoimmune disease and how might it develop?
2. What is an immunologically privileged site and how is it related to transplantation success?
3. How does a tissue rejection reaction occur?
4. Describe an immunodeficiency. How might immunodeficiencies arise?

33.3 Antigen-Antibody Interactions In Vitro

In the previous chapter, section 32.6 (*see pp. 756–57*) deals with those antigen-antibody interactions that occur within an animal (in vivo). Many of these same reactions can take place outside the animal (in vitro) under controlled laboratory conditions and are extensively used in diagnostic testing. The branch of medical immunology concerned with antigen-antibody reactions in vitro is **serology** [serum and -ology].

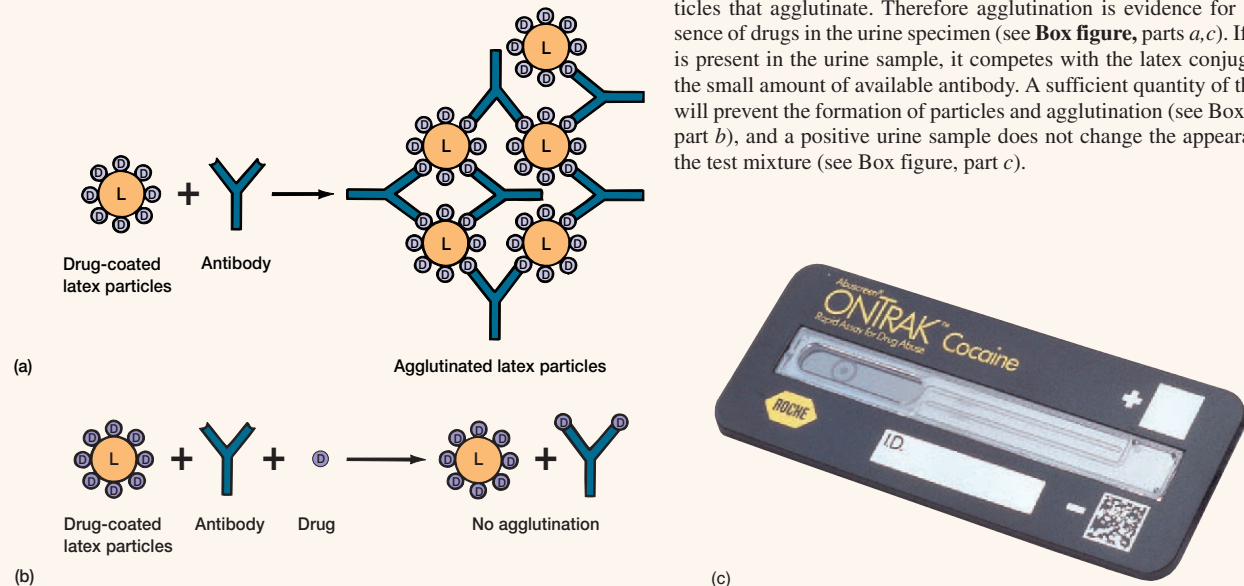
Box 33.2

The Rapid Detection of Drugs in Urine

Historically, drug-screening assays were done by classical chemical methods such as thin-layer chromatography (TLC) or liquid chromatography (LC) which, though accurate, are laborious procedures. Assays currently employed in limited drug testing are based on antigen-antibody reactions. Some examples are the radioimmunoassay, enzyme immunoassay, and fluorescence immunoassay methods, all of which depend on sophisticated instrumentation. With the requirement for massive drug testing in sports, federal civilian employment, the military, and other fields, rapid procedures bypassing the need for expensive equipment are needed.

One such rapid procedure uses a latex agglutination immunoassay for the detection of cocaine, morphine, barbiturates, THC (marijuana), methadone, pencyclidine, and amphetamines. This method provides the accuracy of the immunoassay approach without the need for expensive equipment, and gives accurate on-site “yes” and “no” results within 3 minutes.

The latex agglutination-inhibition test relies on competition for the antibody between a latex-drug conjugate and any drug that may be present in the urine. A urine sample is placed in the mixing well of a slide containing antibody reagent, buffer, and latex reagent. If the drug is absent, the latex-drug conjugate binds to the antibody and forms large particles that agglutinate. Therefore agglutination is evidence for the absence of drugs in the urine specimen (see **Box figure**, parts *a,c*). If a drug is present in the urine sample, it competes with the latex conjugate for the small amount of available antibody. A sufficient quantity of the drug will prevent the formation of particles and agglutination (see **Box figure**, part *b*), and a positive urine sample does not change the appearance of the test mixture (see **Box figure**, part *c*).



Rapid Urine Testing for Drugs. (a) An illustration of a reaction with a negative urine sample. (b) A positive urine sample. (c) One rapid test for cocaine is called Abuscreen and is manufactured by Roche Diagnostic Systems.

As medical immunology enters the new millennium, there has been a marked increase in the number, sensitivity, and specificity of serological tests. The increase results from a better understanding of the cell surface of various lymphocytes, the production of monoclonal antibodies, the development of radioactive and enzyme-linked assays, and the use of fluorescence technology. In this section some more common serological tests employed in the diagnosis of microbial and immunologic diseases are presented.

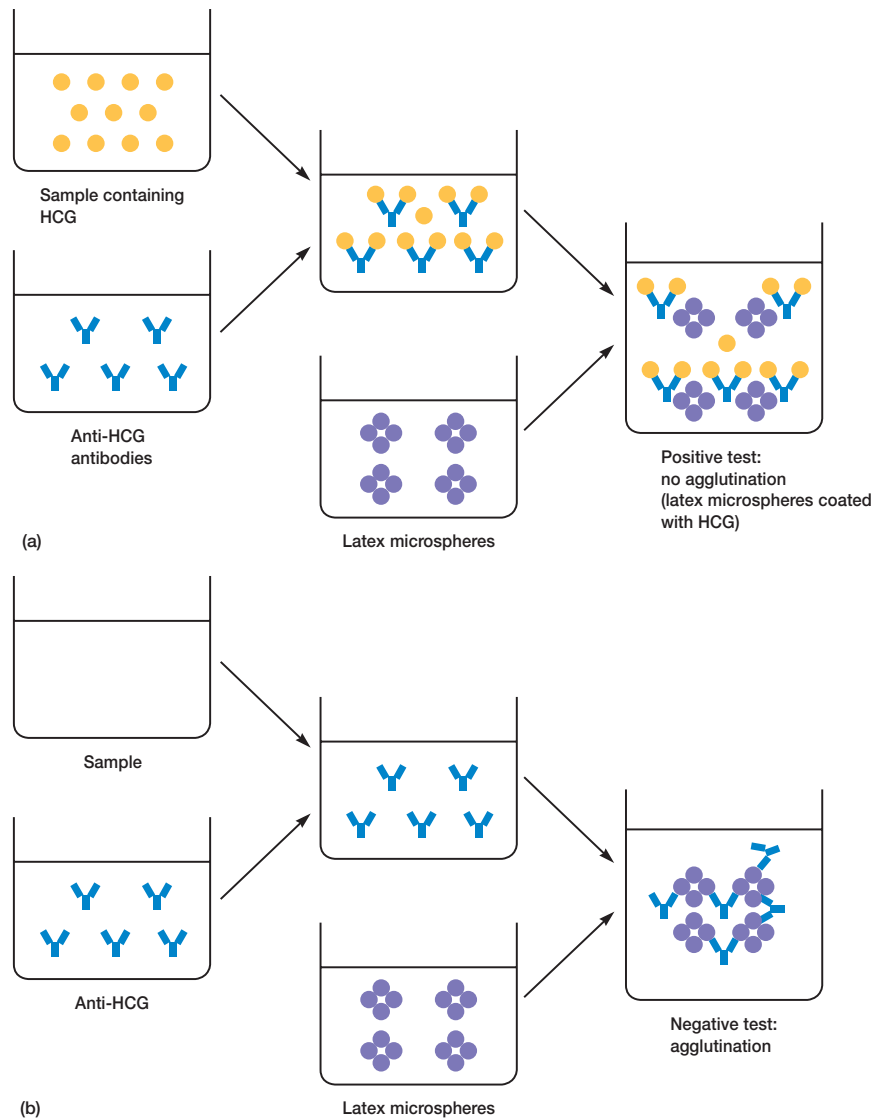
Agglutination

As noted in figure 32.30, when an immune complex is formed by cross-linking cells or particles with specific antibodies, it is called an agglutination reaction. Agglutination reactions usually form vis-

ible aggregates or clumps (**agglutinates**) that can be seen with the unaided eye. Direct agglutination reactions are very useful in the diagnosis of certain diseases. For example, the **Widal test** is a reaction involving the agglutination of typhoid bacilli when they are mixed with serum containing typhoid antibodies from an individual who has typhoid fever. [Immune complex formation \(pp. 756–57\)](#)

Techniques have also been developed that employ microscopic synthetic latex spheres coated with antigens. These coated microspheres are extremely useful in diagnostic agglutination reactions. For example, the modern pregnancy test detects the elevated level of human chorionic gonadotropin (HCG) hormone that occurs in female urine and blood early in pregnancy (**figure 33.9**). Latex agglutination tests are also used to detect antibodies that develop during certain mycotic, helminthic, and bacterial infections, and in drug testing (**Box 33.2**).

Figure 33.9 Latex Agglutination Test for Pregnancy. (a) In a positive test, urine from the female containing HCG is mixed with a solution of antibody specific for HCG. In the second step latex microspheres coated with HCG are added. If HCG is present, it binds to HCG-specific antibodies, thereby preventing them from agglutinating the microspheres. (b) In a negative test the microspheres coated with antigen are agglutinated by HCG-specific antibody.



Hemagglutination usually results from antibodies cross-linking red blood cells through attachment to surface antigens and is routinely used in blood typing. In addition, some viruses can accomplish **viral hemagglutination**. For example, if a person has a certain viral disease, such as measles, antibodies will be present in the serum to react with the measles viruses and neutralize them. Normally, hemagglutination occurs when measles viruses and red blood cells are mixed but is not seen when the person's serum is added to the mixture; this shows that the serum antibodies have neutralized the measles viruses and is considered a positive test (**figure 33.10**). This hemagglutination inhibition test is widely used to diagnose influenza, measles, mumps, mononucleosis, and other viral infections (*see chapter 38*).

Agglutination tests are also used to measure antibody titer (*see figure 32.19*). In the tube or well agglutination test, a spe-

cific amount of antigen is added to a series of tubes (**figure 33.11a**) or shallow wells in a microtiter plate (**figure 33.11b**). Serial dilutions of serum (1/20, 1/40, 1/80, 1/160, etc.) containing the antibody are then added to each tube or well. The greatest dilution of serum showing an agglutination reaction is determined, and the reciprocal of this dilution is the serum antibody titer.

1. What is serology?
2. When would you use the Widal test?
3. Why does hemagglutination occur and how can it be used in the clinical laboratory?
4. Describe how latex agglutination tests work.

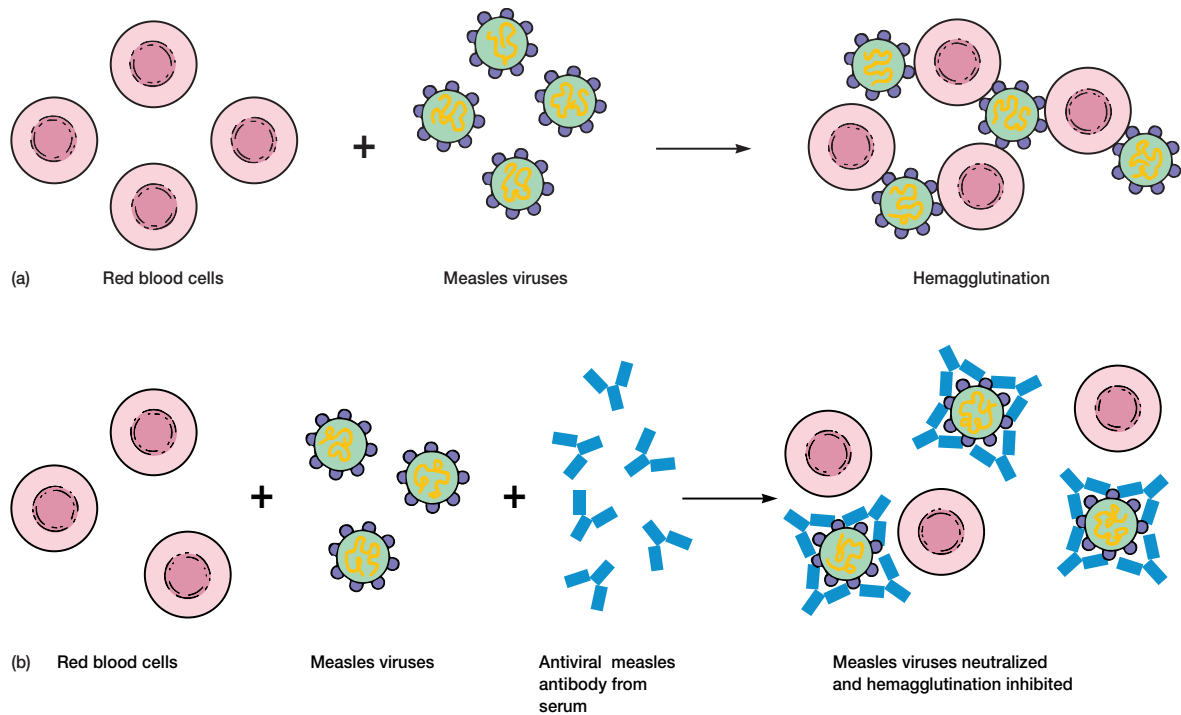


Figure 33.10 Viral Hemagglutination. (a) Certain viruses can bind to red blood cells causing hemagglutination. (b) If serum containing specific antibodies to the virus is mixed with the red blood cells, the antibodies will neutralize the virus and inhibit hemagglutination (a positive test).

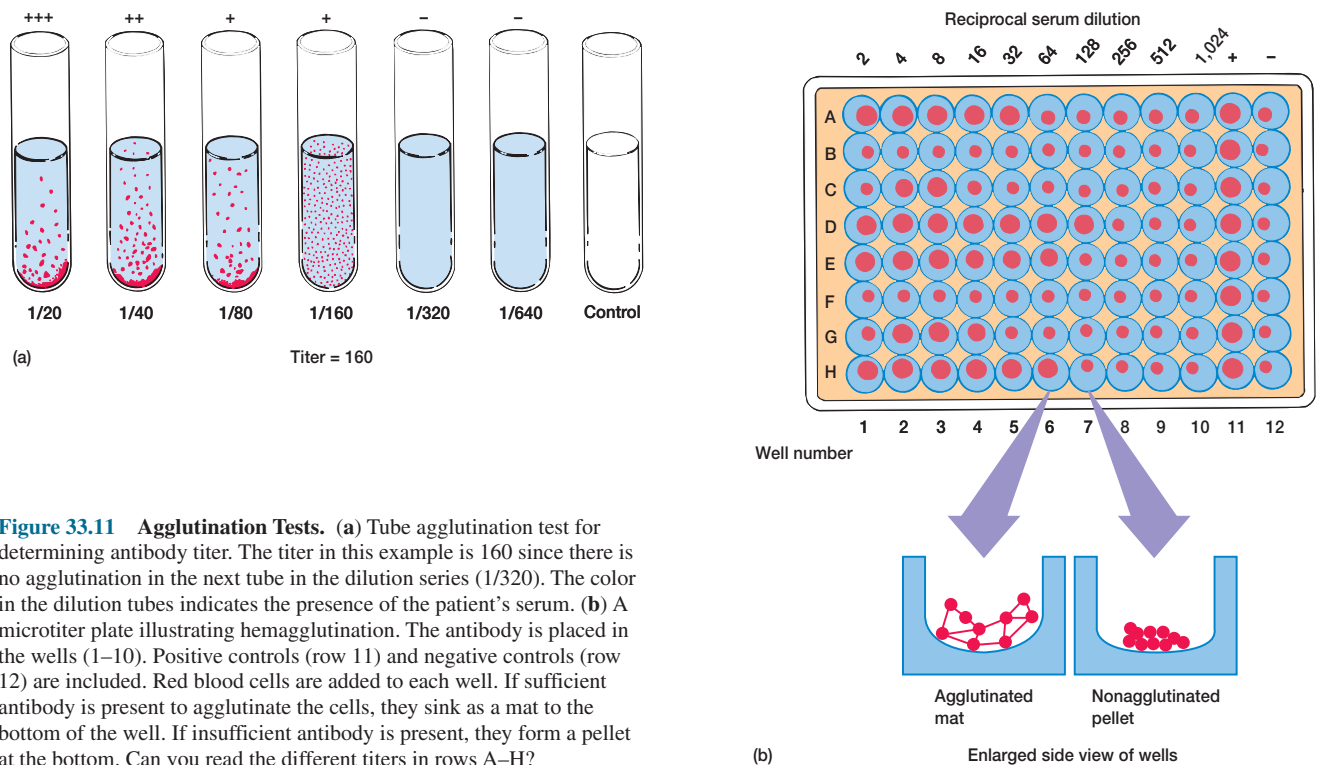


Figure 33.11 Agglutination Tests. (a) Tube agglutination test for determining antibody titer. The titer in this example is 160 since there is no agglutination in the next tube in the dilution series (1/320). The color in the dilution tubes indicates the presence of the patient's serum. (b) A microtiter plate illustrating hemagglutination. The antibody is placed in the wells (1–10). Positive controls (row 11) and negative controls (row 12) are included. Red blood cells are added to each well. If sufficient antibody is present to agglutinate the cells, they sink as a mat to the bottom of the well. If insufficient antibody is present, they form a pellet at the bottom. Can you read the different titers in rows A–H?

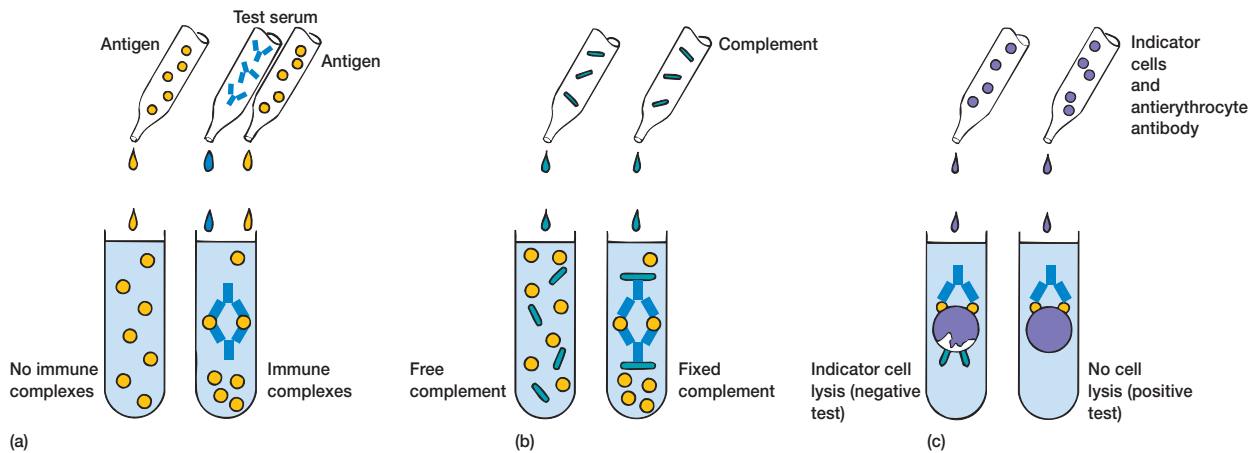


Figure 33.12 Complement Fixation. (a) Test serum is added to one test tube. A fixed amount of antigen is then added to both tubes. If antibody is present in the test serum, immune complexes form. (b) When complement is added, if complexes are present, they fix complement and consume it. (c) Indicator cells and a small amount of antierythrocyte antibody are added to the two tubes. If there is complement present, the indicator cells will lyse (a negative test); if the complement is consumed, no lysis will occur (a positive test).

Complement Fixation

When complement binds to an antigen-antibody complex, it becomes “fixed” and “used up.” Complement fixation tests are very sensitive and can be used to detect extremely small amounts of an antibody for a suspect microorganism in an individual’s serum. A known antigen is mixed with test serum lacking complement (figure 33.12a). When immune complexes have had time to form, complement is added (figure 33.12b) to the mixture. If immune complexes are present, they will fix and consume complement. Afterward, sensitized indicator cells, usually sheep red blood cells previously coated with complement-fixing antibodies, are added to the mixture. Lysis of the indicator cells (figure 33.12c) results if immune complexes do not form in part a of the test because the antibodies are not present in the test serum. In the absence of antibodies, complement remains and lyses the indicator cells. On the other hand, if the specific antibodies are present in the test serum and complement is consumed by the immune complexes, insufficient amounts of complement will be available to lyse the indicator cells. Absence of lysis shows that specific antibodies are present in the test serum.

Complement fixation was once used in the diagnosis of syphilis (the Wassermann test) and is currently used in the diagnosis of certain viral, fungal, rickettsial, chlamydial, and protozoan diseases.

Enzyme-Linked Immunosorbent Assay

The **enzyme-linked immunosorbent assay (ELISA)** has become one of the most widely used serological tests for antibody or antigen detection. This test involves the linking of various “label” enzymes to either antigens or antibodies. Two basic methods are used: the double antibody sandwich assay and the indirect immunosorbent assay.

The double antibody sandwich assay is used for the detection of antigens (figure 33.13a). In this assay, specific antibody is placed in wells of a microtiter plate (or it may be attached to a membrane). The antibody is absorbed onto the walls, coating and sensitizing the plate. A test antigen is then added to each well. If the antigen reacts with the antibody, the antigen is retained when the well is washed to remove unbound antigen. An antibody-enzyme conjugate specific for the antigen is then added to each well. The final complex is formed of an outer antibody-enzyme, middle antigen, and inner antibody—that is, it is a layered (Ab-Ag-Ab) sandwich. A substrate that the enzyme will convert to a colored product is then added, and any resulting product is quantitatively measured by optical density scanning of the plate. If the antigen has reacted with the absorbed antibodies in the first step, the ELISA test is positive. If the antigen is not recognized by the absorbed antibody, the ELISA test is negative because the unattached antigen has been washed away, and no antibody-enzyme is bound. This assay is currently being used for the detection of *Helicobacter pylori* infections, brucellosis, salmonellosis, and cholera. Many other antigens also can be detected by the sandwich method. For example, there are ELISA kits on the market that can test for many different food allergens.

The indirect immunosorbent assay detects antibodies rather than antigens. In this assay, antigen in appropriate buffer is incubated in the wells of a microtiter plate (figure 33.13b) and is absorbed onto the walls of the wells. Free antigen is washed away. Test antiserum is added, and if specific antibody is present, it binds to the antigen. Unbound antibody is washed away. Alternatively the test sample can be incubated with a suspension of latex beads that have the desired antigen attached to their surface. After allowing time for antibody-antigen complex formation, the beads are trapped on a filter and unbound antibody is washed

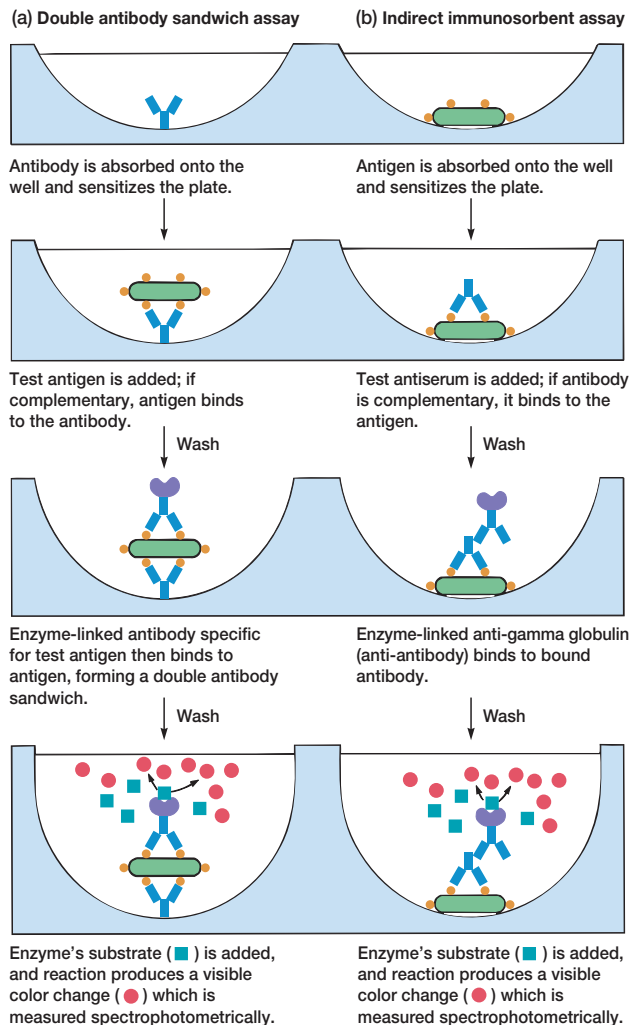


Figure 33.13 The ELISA or EIA Test. (a) The double antibody sandwich method for the detection of antigens. (b) The indirect immunosorbent assay for detecting antibodies. See text for details.

away. An anti-antibody that has been covalently coupled to an enzyme, such as horseradish peroxidase, is added next. The antibody-enzyme complex (the conjugate) binds to the test antibody, and after unbound conjugate is washed away, the attached ligand is visualized by the addition of a chromogen. A **chromogen** is a colorless substrate acted on by the enzyme portion of the ligand to produce a colored product. The amount of test antibody is quantitated in the same way as an antigen is in the double antibody sandwich method. The indirect immunosorbent assay currently is being used to test for antibodies to human immunodeficiency virus (the causative agent of AIDS) and rubella virus (German measles), and to detect certain drugs in serum. For example, antigen-coated latex beads are used in the SUDS HIV-1 test to detect HIV serum antibodies in about 10 minutes.

Flow Cytometry and Fluorescence

Classical microbiological techniques (*see section 36.2*) are relatively slow in comparison to analytical techniques, in many cases due to the need to culture microorganisms. Furthermore, classical techniques are not useful in identifying those microorganisms that can't be cultured. The recent emergence of molecular biological techniques, particularly those using antibodies and nucleic acid probes combined with amplification techniques, has provided the clinical laboratory with rapid and very specific techniques for microbiological diagnosis. One such new technique is flow cytometry and fluorescence.

Flow cytometry allows single- or multiple-microorganism detection in clinical samples (*see section 36.1*) in an easy, reliable, fast way. In flow cytometry, microorganisms are identified on the basis of their peculiar cytometric parameters or by means of fluorochromes that can be used either independently or bound to specific antibodies or oligonucleotides. The flow cytometer forces a suspension of cells through a laser beam and measures the light they scatter or the fluorescence the cells emit as they pass through the beam. For example, cells can be tagged with a fluorescent antibody directed against a specific surface antigen. As the stream of cells flows past the laser beam, the fluorescent cells can be detected, counted, and even separated from the other cells in the suspension. The cytometer also can measure a cell's shape, size, and content of DNA and RNA. This technique has permitted the development of quantitative methods to access antimicrobial susceptibility and drug cytotoxicity in a rapid, accurate, and highly reproducible way. The most outstanding contribution of this technique is the ability to detect the presence of heterogeneous microbial populations with different responses to antimicrobial treatments.

Immunoblotting (Western Blot)

Another immunologic technique used in the clinical microbiology laboratory is **immunoblotting**. Immunoblotting involves polyacrylamide gel electrophoresis of a protein specimen followed by transfer of the separated proteins to nitrocellulose sheets (*see figure 14.5*). Protein bands are then visualized by treating the nitrocellulose sheets with solutions of enzyme-tagged antibodies. This procedure demonstrates the presence of common and specific proteins among different strains of microorganisms (*figure 33.14*). Immunoblotting also can be used to show strain-specific immune responses to microorganisms, to serve as an important diagnostic indicator of a recent infection with a particular strain of microorganism, and to allow for prognostic implications with severe infectious diseases.

Immunodiffusion

Immunodiffusion refers to a precipitation reaction that occurs between an antibody and antigen in an agar gel medium. Two techniques are routinely used: single radial immunodiffusion and double diffusion in agar.

The **single radial immunodiffusion (RID) assay** or Mancini technique quantitates antigens. Monospecific antibody is added to agar, then the mixture is poured onto slides and allowed to set.

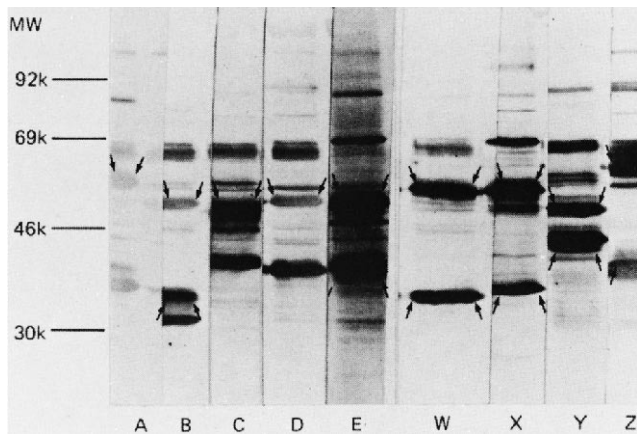
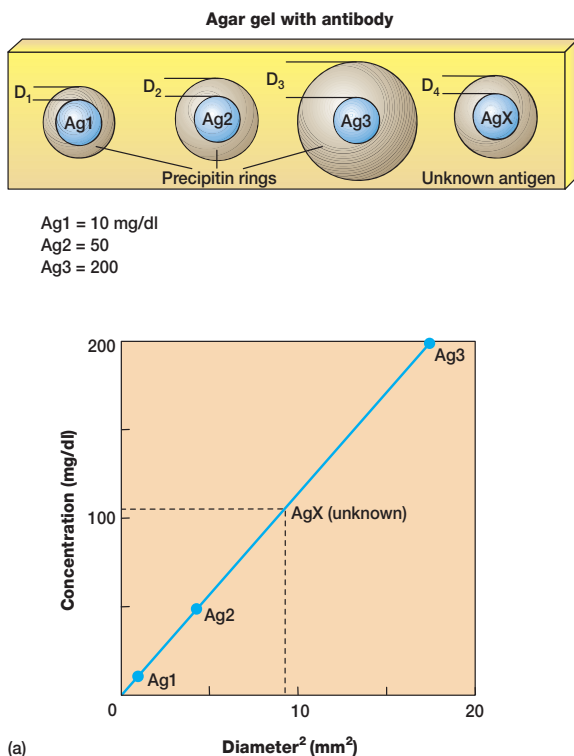


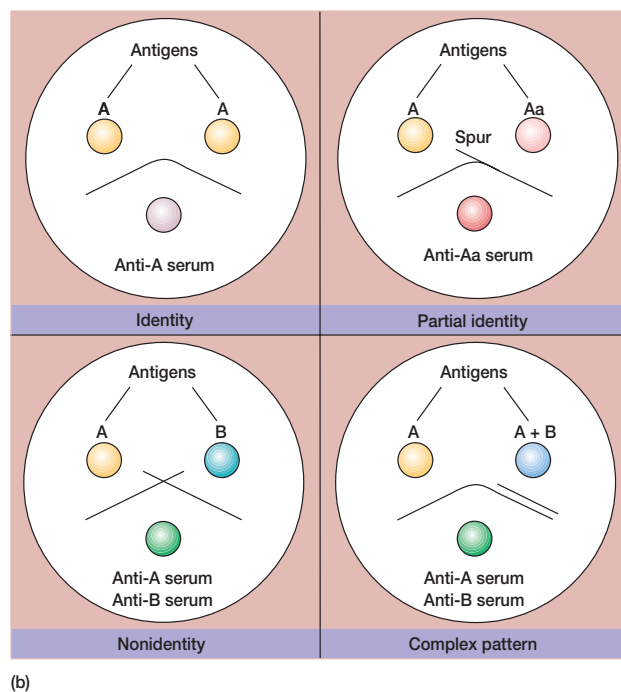
Figure 33.14 Immunoblotting (Western Blot). Immunoblot of the standard strains of *Clostridium difficile*. Arrows indicate strain-specific bands (specific molecular weights) of the various proteins in different lanes (A–E). The molecular weight of the protein is indicated on the left.

Wells are cut in the agar and known amounts of standard antigen added. The unknown test antigen is added to a separate well (figure 33.15a). The slide is left for 24 hours or until equilibrium has been reached, during which time the antigen diffuses out of the wells to form insoluble complexes. The size of the resulting precipitation ring surrounding various dilutions of antigen selected is proportional to the amount of antigen in the well (the wider the ring, the greater the antigen concentration). This is because the antigen's concentration drops as it diffuses farther out into the agar. The antigen forms a precipitin ring in the agar when its level has decreased sufficiently to reach equivalence and combine with the antibody to produce a large, insoluble network. This method is commonly used to quantitate serum immunoglobulins, complement proteins, and other substances.

The **double diffusion agar assay (Ouchterlony technique)** is based on the principle that diffusion of both antibody and antigen (hence, double diffusion) through agar can form stable and easily observable immune complexes. Test solutions of antigen and antibody are added to the separate wells punched in agar. The solutions diffuse outward, and when antigen and the appropriate antibody meet, they combine and precipitate at the equivalence zone, producing an indicator line (or lines) (figure 33.15b). The visible



(a)



(b)

Figure 33.15 Immunodiffusion. (a) Single radial immunodiffusion assay. Three standard solutions of different antigen concentrations (Ag1, Ag2, Ag3) and an unknown (AgX) are placed on agar. After equilibration the ring diameters are measured. Usually the square of the diameter of the standard rings is plotted on the x-axis and the antigen concentration on the y-axis. From this standard curve the concentration of an unknown can be determined. (b) Double diffusion agar assay showing characteristics of identity (top left), reaction of nonidentity (bottom left), partial identity (top right), and a complex pattern (bottom right).

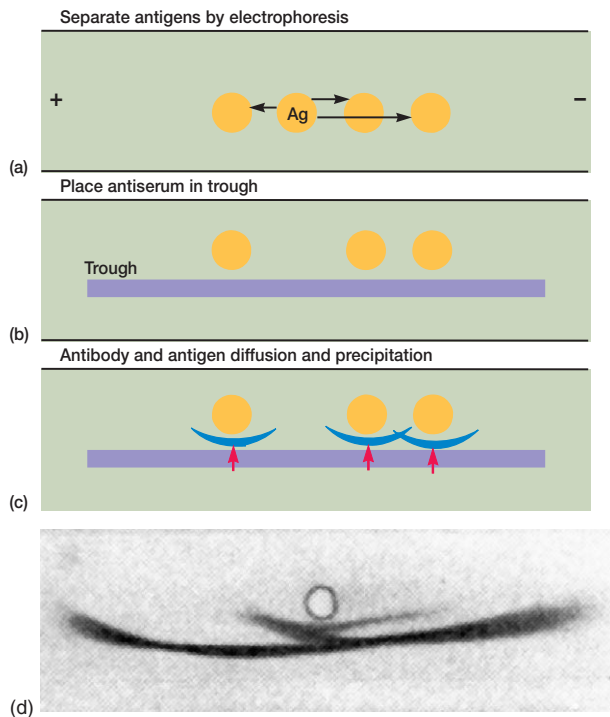


Figure 33.16 Classical Immunoelectrophoresis. (a) Antigens are separated in an agar gel by an electrical charge. (b) Antibody (antiserum) is then placed in a trough cut parallel to the direction of the antigen migration. (c) The antigens and antibodies diffuse through the agar and form precipitin arcs. (d) After staining, better visualization is possible.

line of precipitation permits a comparison of antigens for identity (same antigenic determinants), partial identity (cross-reactivity), or nonidentity against a given selected antibody. For example, if a V-shaped line of precipitation forms, this demonstrates that the antibodies bind to the same antigenic determinants in each antigen sample and are identical. If one well is filled with a different antigen that shares some but not all determinants with the first antigen, a Y-shaped line of precipitation forms, demonstrating partial identity. In this reaction the stem of the Y, called a spur, is formed if those antigen or antigenic determinants absent in the first well but present in the second one (antigen a in figure 33.15b) react with the diffusing antibodies. If two completely unrelated antigens are added to the wells, either a single straight line of precipitation forms between the two wells, or two separate lines of precipitation form, creating an X-shaped pattern, a reaction of nonidentity.

Immunoelectrophoresis

Some antigen mixtures are too complex to be resolved by simple diffusion and precipitation. Greater resolution is obtained by the technique of classical **immunoelectrophoresis** in which antigens are first separated based on their electrical charge, then visualized by the precipitation reaction. In this procedure antigens are separated by electrophoresis in an agar gel. Positively charged pro-

teins move to the negative electrode, and negatively charged proteins move to the positive electrode (**figure 33.16a**). A trough is then cut next to the wells (**figure 33.16b**) and filled with antibody. If the plate is incubated, the antibodies and antigens will diffuse and eventually form precipitation bands or arcs (**figure 33.16c**) that can be better visualized by staining (**figure 33.16d**). This assay is used to separate the major blood proteins in serum for certain diagnostic tests. [Electrophoresis \(p. 327\)](#)

Immunofluorescence

Immunofluorescence is a process in which dyes called fluorochromes are exposed to UV, violet, or blue light to make them fluoresce or emit visible light. Dyes such as rhodamine B or fluorescein isothiocyanate (FITC) can be coupled to antibody molecules without changing the antibody's capacity to bind to a specific antigen. Fluorochromes also can be attached to antigens. There are two main kinds of fluorescent antibody assays: direct and indirect.

Direct immunofluorescence involves fixing the specimen (cell or microorganism) containing the antigen of interest onto a slide (**figure 33.17a**). Fluorescein-labeled antibodies are then added to the slide and incubated. The slide is washed to remove any unbound antibody and examined with the fluorescence microscope (*see figure 2.12*) for a yellow-green fluorescence. The pattern of fluorescence reveals the antigen's location. Direct immunofluorescence is used to identify antigens such as those found on the surface of group A streptococci and to diagnose enteropathogenic *Escherichia coli*, *Neisseria meningitidis*, *Salmonella typhi*, *Shigella sonnei*, *Listeria monocytogenes*, *Haemophilus influenzae* type b, and the rabies virus. [The fluorescence microscope \(pp. 25–26\)](#)

Indirect immunofluorescence (**figure 33.17b**) is used to detect the presence of antibodies in serum following an individual's exposure to microorganisms. In this technique a known antigen is fixed onto a slide. The test antiserum is then added, and if the specific antibody is present, it reacts with antigen to form a complex. When fluorescein-labeled anti-immunoglobulin is added, it reacts with the fixed antibody. After incubation and washing, the slide is examined with the fluorescence microscope. The occurrence of fluorescence shows that antibody specific to the test antigen is present in the serum. Indirect immunofluorescence is used to identify the presence of *Treponema pallidum* antibodies in the diagnosis of syphilis (treponemal antibody absorption, FTA-ABS; *see figure 39.21*) as well as antibodies produced in response to other microorganisms.

Immunoprecipitation

The **immunoprecipitation** technique detects soluble antigens that react with antibodies called precipitins. The precipitin reaction occurs when bivalent or multivalent antibodies and antigens are mixed in the proper proportions. The antibodies link the antigen to form a large antibody-antigen network or lattice that settles out of solution when it becomes sufficiently large (**figure 33.18a**). Immunoprecipitation reactions occur only at the equivalence zone when there is an optimal ratio of antigen to antibody so that a lattice forms. If the precipitin reaction takes place in a test tube (**figure 33.18b**), a precipitation ring forms in the area in which the optimal ratio or equivalence zone develops. [Immune complex formation \(p. 756\)](#)

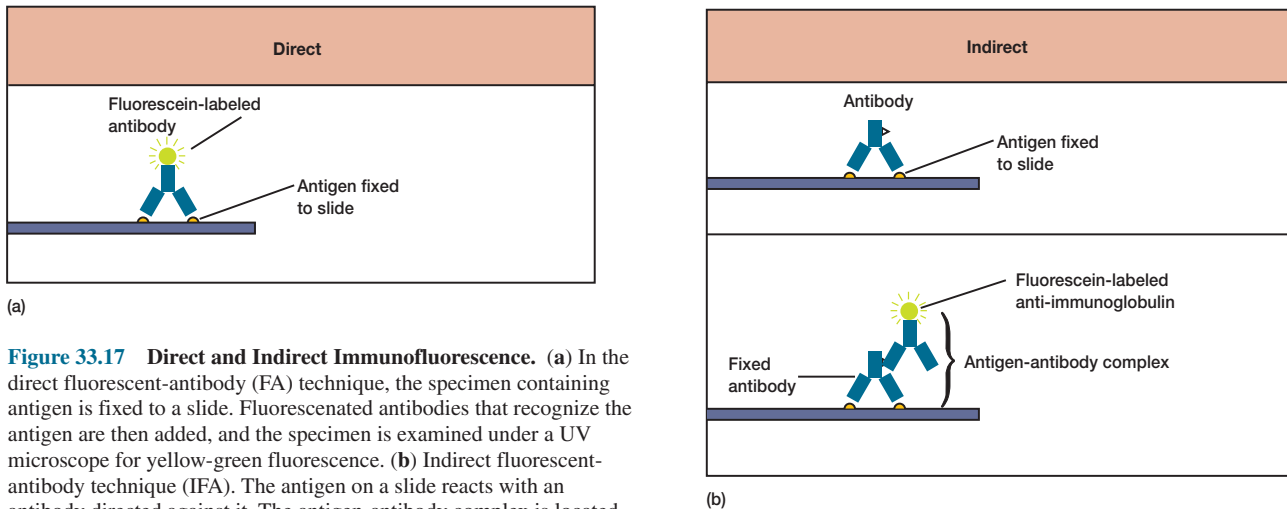


Figure 33.17 Direct and Indirect Immunofluorescence. (a) In the direct fluorescent-antibody (FA) technique, the specimen containing antigen is fixed to a slide. Fluorescent antibodies that recognize the antigen are then added, and the specimen is examined under a UV microscope for yellow-green fluorescence. (b) Indirect fluorescent-antibody technique (IFA). The antigen on a slide reacts with an antibody directed against it. The antigen-antibody complex is located with a fluorescent antibody that recognizes immunoglobulins.

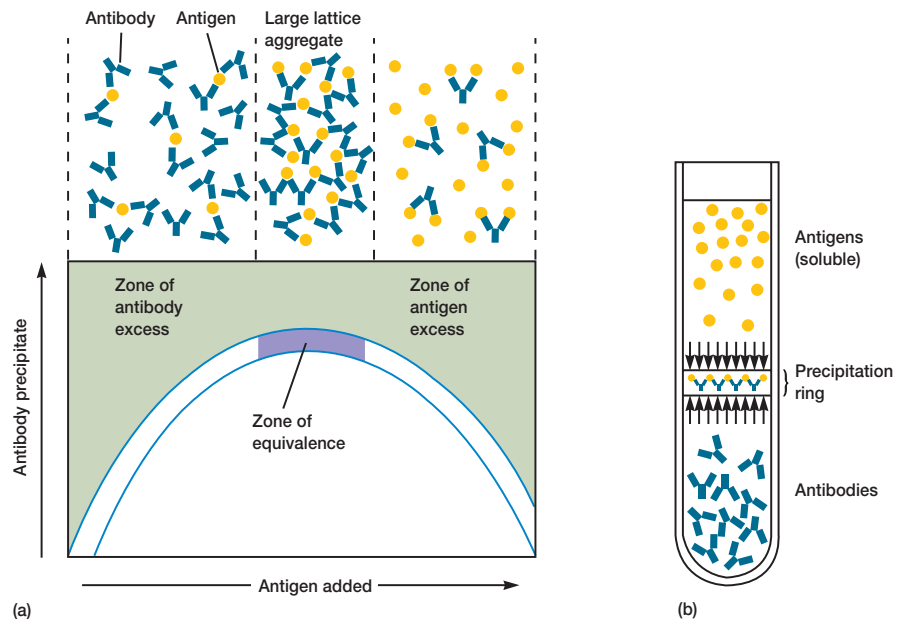


Figure 33.18 Immunoprecipitation. (a) Graph showing that a precipitation curve is based on the ratio of antigen to antibody. The zone of equivalence represents the optimal ratio for precipitation. (b) A precipitation ring test. Antibodies and antigens diffuse toward each other in a test tube. A precipitation ring is formed at the zone of equivalence.

1. What does a negative complement fixation test show? A positive test?
2. What are the two types of ELISA methods and how do they work? What is a chromogen?
3. What are some advantages of using flow cytometry in comparison to classical microbiological techniques to detect microorganisms?
4. Name two types of immunodiffusion tests and describe how they operate.
5. Describe the classical immunoelectrophoresis technique.
6. What is a fluorochrome?
7. Name and describe the two kinds of fluorescent antibody assays.
8. Specifically, when do immunoprecipitation reactions occur?

Liposomes

Another immunologic technique uses liposomes. A **liposome** [Greek *lipos*, fat, and *soma*, body] is an artificially created microscopic spherical vesicle formed by a lipid bilayer enclosing an aqueous compartment. The aqueous compartment contains a colored dye. The liposome is then sensitized by coupling a specific antibody (or antigen) to the liposome surface. Antibodies specific for the pathogen to be detected are attached to a membrane in a particular shape, such as a triangle. After a patient's sample is added to the test well (**figure 33.19a**), the substance undergoing analysis instantly binds to the antibodies (or antigen). When the dye-filled sensitized liposomes are added (**figure 33.19b**) and link to the immobilized substance (**figure 33.19c**), a triangle (or other

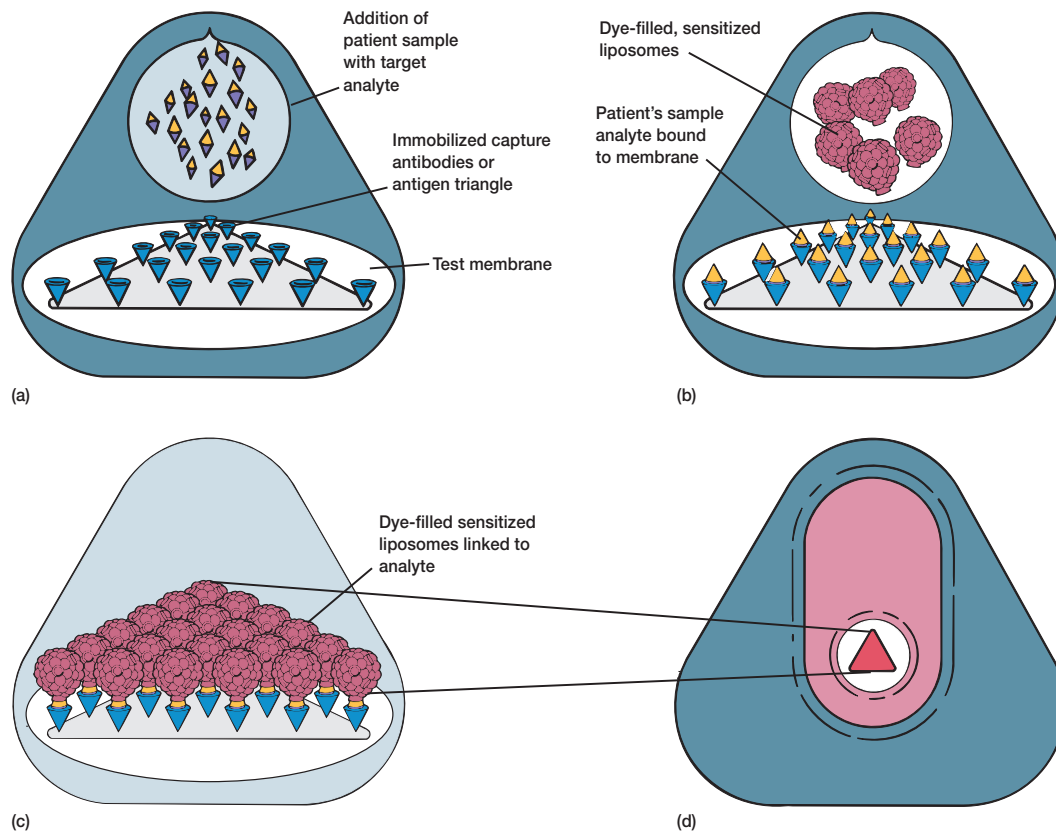


Figure 33.19 The Use of Liposomes in Diagnostic Testing. (a) The addition of a patient's sample containing the target analyte. Immobilized capture antibodies or antigen on a test membrane form a geometric design such as a triangle. (b) When dye-filled liposomes are added, they bind to the patient's bound analyte (c) giving a triangle. (d) The triangle indicates a positive reaction.

shape) indicates a positive reaction (figure 33.19d). If the sample is negative and contains none of the substance, no binding occurs in the test area when the liposomes are added and either a blank or some other shape appears. Liposome tests are currently available for group A streptococci and the respiratory syncytial virus.

Neutralization

Neutralization tests are antigen-antibody reactions that determine whether the activity of a toxin or virus has been neutralized by antibody. Laboratory animals or tissue culture cells are used as "indicator systems" in these tests. The toxin or virus to be assayed has known effects on the indicator system. The effect in the animal might be death, paralysis, or skin lesions. For example, when the exotoxin of *Clostridium botulinum* is suspected of causing food poisoning in a person (see section 39.4), a sample of either the suspect food or the serum, stool, or vomitus of the ill patient is collected. Two groups of indicator mice are used. The control group receives the botulinum antitoxin and the experimental group does not. Filtrates from the source samples are injected into both groups of mice. If toxin is present, all mice except those receiving the antitoxin will die, and the test is positive for food botulism.

Viral neutralization assays are frequently used to detect viral infections. Suspected blood serum containing viral antibodies can be introduced into tissue culture cells or embryonated eggs (see figure 16.1). If antibodies are present against the virus, viral neutralization will occur and prevent the virus from infecting the culture cells. No cytopathic effects will be seen.

Radioimmunoassay

The **radioimmunoassay (RIA)** technique has become an extremely important tool in biomedical research and clinical practice (e.g., in cardiology, blood banking, diagnosis of allergies, and endocrinology). Indeed, Rosalyn Yalow won the 1977 Nobel Prize in Physiology or Medicine for its development. RIA uses a purified antigen that is radioisotope-labeled and competes for antibody with unlabeled standard or antigen in experimental samples. The radioactivity associated with the antibody is then detected by means of radioisotope analyzers and autoradiography (photographic emulsions that show areas of radioactivity). If there is much antigen in an experimental sample, it will compete with the radioisotope-labeled antigen for antigen-binding sites on the antibody, and little radioactivity will be bound. A large amount of

Box 33.3

History and Importance of Serotyping

In the early 1930s Rebecca Lancefield (1895–1981) recognized the importance of serological tests. She developed a classification system for the streptococci based on the antigenic nature of cell wall carbohydrates. Her system is now known as the **Lancefield system** in which each different serotype is a Lancefield group and identified by a letter (A through T). This scheme is based on specific antibody agglutination reactions with cell wall carbohydrate antigens (C polysaccharides) extracted from the streptococci. Lancefield also showed that further subdividing of the group A streptococci into specific serological types was possible, based on the presence of type-specific M (protein) antigens.

More recently *Escherichia coli*, *Salmonella* and other bacteria have been serotyped with specific antigen-antibody reactions involving flagella (H) antigens, capsular (K) antigens, and somatic (cell wall or O) antigens. Within *E. coli* there are over 167 different O antigens.

The current value of serotyping may be seen in the fact that *E. coli* O55, O111, and O127 serotypes are the ones most frequently associated with infantile diarrhea. Thus the serotype of *E. coli* from stool samples of infants with infantile diarrhea is of diagnostic value and aids in identifying the source of the infection.

bound radioactivity indicates that there is little antigen present in the experimental sample.

Serotyping

Serotyping refers to serological procedures used to differentiate strains (serovars or serotypes) of microorganisms that differ in the antigenic composition of a structure or product (**Box 33.3**). The serological identification of a pathogen strain has diagnostic value. Often the symptoms of infections depend on the nature of cell products released by a pathogen. Genes for virulence often occur in the same clone with genes for antigenic cell wall material. Therefore it is possible to identify a pathogen serologically by testing for cell wall antigens. For example, there are 84 strains of *Streptococcus pneumoniae*, each differing in the nature of its capsular material. These differences can be detected by capsular swelling (termed the **Quellung reaction**) if antisera specific for the capsular types are used (**figure 33.20**).

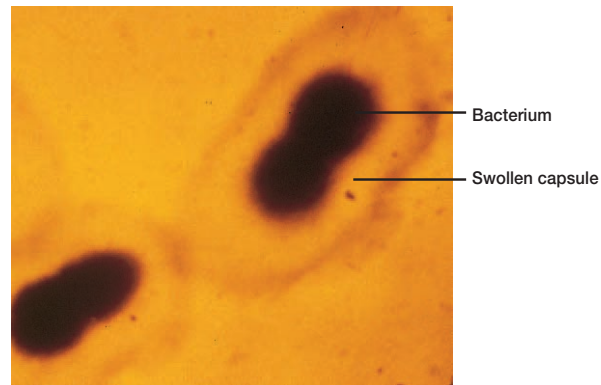


Figure 33.20 Serotyping. *Streptococcus pneumoniae* has reacted with a specific pneumococcal antiserum leading to capsular swelling (the Quellung reaction). The capsules seen around the bacteria indicate potential virulence.

1. What is a neutralization test?
2. Describe the RIA technique. What is serotyping?
3. How are liposomes used to test for the respiratory syncytial virus?

Summary

1. Vaccination is one of the most cost-effective weapons for microbial disease prevention, and vaccines constitute one of the greatest achievements of modern medicine.
2. Artificial immunity to pathogens can be accomplished by either active or passive immunization (**tables 33.1, 33.2 and figure 33.1**).
3. Many of the current vaccines in use for humans (**table 33.3**) consists of whole organisms that are either inactivated (killed) or attenuated (live but avirulent).
4. Some of the risks associated with whole-organism vaccines can be avoided by using only specific purified macromolecules derived from pathogenic microorganisms
5. Recently a number of microorganisms have been used for recombinant-vector vaccines. The attenuated microorganism serves as a

(**table 33.4**). Currently, there are three general forms of macromolecule vaccines: capsular polysaccharides, recombinant surface antigens, and inactivated exotoxins (toxoids).

vector, replicating within the host, and expressing the gene product of the pathogen-encoded antigenic proteins. The proteins can elicit humoral immunity when the proteins escape from the cells and cellular immunity when they are broken down and properly displayed on the cell surface.

6. More complicated genetic vaccines are termed DNA vaccines. DNA vaccines elicit protective immunity against a pathogen by activating both branches of the immune system: humoral and cellular.
7. When the immune response occurs in an exaggerated form and results in tissue damage to the individual, the term hypersensitivity is applied. There are four types of hypersensitivity reactions, designated as types I through IV (**figures 33.2–33.7**).
8. Autoimmune diseases result when self-reactive T and B cells attack the body and cause tissue damage (**table 33.5**). A variety of factors can influence the development of autoimmune disease.
9. The immune system can act detrimentally and reject tissue transplants. There are different types of transplants. Xenografts involve transplants of privileged tissue between different species, and allografts are transplants between genetically different individuals of the same species.
10. Immunodeficiency diseases are a diverse group of conditions in which an individual's susceptibility to various infections is increased; several severe diseases can arise because of one or more defects in the specific or nonspecific immune response.
11. Agglutination reactions in vitro usually form aggregates or clumps (agglutinates) that are visible with the naked eye (**figure 33.9**). Tests have been developed, such as the Widal test, latex microsphere agglutination reaction, hemagglutination, and viral hemagglutination (**figure 33.10**), to detect antigen as well as to determine antibody titer.
12. The complement fixation test can be used to detect a specific antibody for a suspect microorganism in an individual's serum (**figure 33.12**).
13. The enzyme-linked immunosorbent assay (ELISA) involves the linking of various enzymes to either antigens or antibodies. Two basic methods are involved: the double antibody sandwich method and the indirect immunosorbent assay (**figure 33.13**). The first method detects antigens and the latter, antibodies.
14. Flow cytometry and fluorescence allow single- or multiple-microorganism detection based on their peculiar cytometric parameters or by means of certain fluorochromes.
15. Immunoblotting involves polyacrylamide gel electrophoresis of a protein specimen followed by transfer of the separated proteins to nitrocellulose sheets and identification of specific bands.
16. Immunodiffusion refers to a precipitation reaction that occurs between antibody and antigen in an agar gel medium. Two techniques are routinely used: double diffusion in agar and single radial diffusion (**figure 33.15**).
17. In classical immunoelectrophoresis antigens are separated based on their electrical charge, then visualized by precipitation and staining (**figure 33.16**).
18. Immunofluorescence is a process in which certain dyes called fluorochromes are irradiated with UV, violet, or blue light to make them fluoresce. These dyes can be coupled to an antibody. There are two main kinds of fluorescent antibody assays: direct and indirect (**figure 33.17**).
19. Immunoprecipitation reactions occur only when there is an optimal ratio of antigen and antibody to produce a lattice at the zone of equivalence, which is evidenced by a visible precipitate (**figure 33.18**).
20. Liposomes are artificially created microscopic spheres that can be used to detect group A streptococci and the respiratory syncytial virus (**figure 33.19**).
21. Neutralization tests are antigen-antibody reactions that determine whether toxin or virus activity has been neutralized by antibody. Laboratory animals or tissue culture cells are used as indicator systems in these tests.
22. Radioimmunoassays use a purified antigen that is radioisotope-labeled and used to compete with unlabeled standard or antigen in experimental samples for a specific antibody.
23. Serotyping refers to serological procedures used to differentiate strains (serovars or serotypes) of microorganisms that have differences in the antigenic composition of a structure or product (**figure 33.20**).

Key Terms

- | | | |
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| active immunization 764 | graft-versus-host disease 773 | serology 774 |
| agglutinates 775 | hay fever 768 | serotyping 784 |
| allergen 768 | hives 769 | single radial immunodiffusion (RID) assay 779 |
| allergic contact dermatitis 771 | hypersensitivity 768 | systemic lupus erythematosus 770 |
| allergy 768 | immunoblotting 779 | TB skin test 771 |
| allograft 773 | immunodeficiencies 774 | toxoid 767 |
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| double diffusion agar assay (Ouchterlony technique) 780 | Quellung reaction 784 | Widal test 775 |
| enzyme-linked immunosorbent assay (ELISA) 778 | radioimmunoassay (RIA) 783 | xenograph 773 |
| flow cytometry 779 | reagin 768 | |
| | recombinant-vector vaccine 767 | |

Questions for Thought and Review

1. Describe how IgE can have both beneficial and detrimental effects in the same host. Why have its detrimental effects persisted? Explain your reasoning.
2. In desensitization procedures, the allergist injects more of the same allergen to which the person is allergic. How can this be beneficial?
3. If excess complement is added in a complement fixation test, how will this affect the results?
4. Specific chemical markers are conjugated to antibodies or antigens in some immunologic tests. What are the advantages of this technique over such tests as agglutination or complement fixation?
5. Of what value are neutralization tests performed in animals or in tissue culture cells?
6. What is the importance of serotyping?
7. What are some features of an in vitro antigen-antibody reaction that makes the reaction so useful for identification and monitoring tests?
8. How are chromogens used as biological markers?
9. Even though *Streptococcus pneumoniae* has over 80 different serotypes, serological reactions are sensitive enough to differentiate one serotype from another. How is this possible?
10. How can electrophoresis be used in immunologic testing procedures?
11. Describe each of the four types of hypersensitivities.
12. Are attenuated vaccines more likely than inactivated vaccines to induce-cell mediated immunity?
13. Do DNA vaccines generate immunological memory? Explain your answer.
14. List the three types of purified molecules that are used as vaccines.
15. Define vaccine immunization.

Critical Thinking Questions

1. How does an inactivated vaccine induce only a humoral response, whereas an attenuated vaccine induces both humoral and cell-mediated responses?
2. Why is a DNA vaccine delivered intramuscularly and not by intravenous or oral routes?
3. ELISA tests usually use a primary and secondary antibody. Why? What are the necessary controls one would need to perform to ensure that the antibody specificity is valid (no false positives or negatives)?

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PART X

Microbial Diseases and Their Control

Chapter 34

Pathogenicity of Microorganisms

Chapter 35

Antimicrobial Chemotherapy

Chapter 36

Clinical Microbiology

Chapter 37

The Epidemiology
of Infectious Disease

Chapter 38

Human Diseases Caused by Viruses

Chapter 39

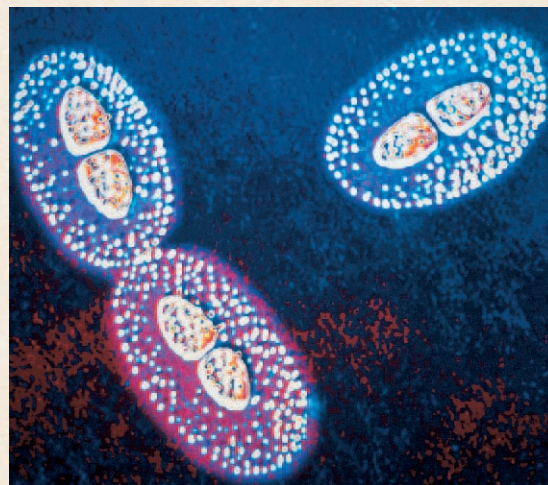
Human Diseases Caused by Bacteria

Chapter 40

Human Diseases Caused by Fungi
and Protozoa

CHAPTER 34

Pathogenicity of Microorganisms



Three *Streptococcus pneumoniae*, each surrounded by a slippery mucoid capsule (shown as a layer of white spheres around the diplococcus bacteria). The polysaccharide capsule is vital to the pathogenicity of this bacterium since it prevents phagocytic cells from accomplishing phagocytosis.

Outline

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Concepts

1. If a microorganism (symbiont) either harms or lives at the expense of another organism, it is called a parasitic organism and the relationship is termed parasitism. In this relationship the body of an animal is referred to as the host.
2. Those organisms capable of causing disease are called pathogens. Disease is any change in the host from a healthy to an unhealthy, abnormal state in which part or all of the host's body is not capable of carrying on its normal functions.
3. The obligatory steps for the infectious process involving viral diseases are: a virus must enter a host, come into contact with susceptible cells, replicate within the cells, spread to adjacent cells, cause cellular injury, engender a host immune response, be cleared from the body of the host or establish a persistent infection, and be shed back into the environment.
4. The obligatory steps for the infectious process involving bacterial diseases are: the bacterium must be transmitted to a suitable host, attach to and/or colonize the host, grow and multiply within or on the host, and interfere with or impair the normal physiological activities of the host.
5. One important result of the conservation of chromosomal genes is that bacteria are clonal. Only one or a few clonal types of some bacterial pathogens exist in the environment.
6. During coevolution with human hosts, some pathogenic bacteria have evolved complex signal transduction pathways to regulate the genes necessary for virulence.
7. Many bacteria are pathogenic because they have large segments of DNA, called pathogenicity islands, that carry genes responsible for virulence.
8. Two distinct categories of disease can be recognized based on the role bacteria play in the disease causing process: infections (invasions) and intoxications.
9. Toxins produced by pathogenic bacteria are either exotoxins or endotoxins.
10. Viruses and bacteria are continuously evolving and producing unique mechanisms that enable them to escape the host's arsenal of defenses.

Pathogenicity is not the rule. Indeed, it occurs so infrequently and involves such a relatively small number of species, considering the huge population of bacteria on earth, that it has a freakish aspect. Disease usually results from inconclusive negotiations for symbiosis, an overstepping of the line by one side or the other, a biological misinterpretation of borders.

—Lewis Thomas

Chapter 28 introduces the concept of symbiosis and deals with two of its subordinate categories: commensalism and mutualism. In this chapter the third category, parasitism, is presented along with one of its possible consequences—pathogenicity. The parasitic way of life is so successful that it has evolved independently in nearly all groups of organisms. In recent years concerted efforts to understand organisms and their relationships with their hosts have developed within the disciplines of virology, rickettsiology, chlamydiology, bacteriology, mycology, parasitology (protozoology and helminthology), entomology, and zoology. This chapter examines the parasitic way of life in terms of health and disease in the animal body and emphasizes viral and bacterial disease mechanisms. The chapter concludes with some viral and bacterial mechanisms used to escape host defenses.

34.1 Host-Parasite Relationships

If a symbiont either harms or lives at the expense of another organism (the **host**), it is a **parasitic organism**, and the relationship is called **parasitism**. In this relationship the body of the host can be viewed as a microenvironment that shelters and supports the growth and multiplication of the parasitic organism. The parasitic organism is usually the smaller of the two partners and is metabolically dependent on the host. There are many parasitic agents or organisms among the viruses, bacteria, fungi, plants, and animals (**table 34.1**). By convention, when the word **parasite** is used without qualification, it refers specifically to a protozoan or helminthic (nematode, trematode, cestode) organism.

Several types of parasitism are recognized. If an organism lives on the surface of its host, it is an **ectoparasite**; if it lives in-

Table 34.1 Categorization of Parasitic Organisms and Agents by Size

Discipline	Parasitic Group	Approximate Size
Virology	Prions	350 kDa
	Viroids	130 kDa
	Viruses	25–400 nm
Bacteriology	Chlamydiae	0.2–1.5 μm
	Mycoplasmas	0.3–0.8 μm
	Rickettsias	0.5–2 μm
	Other bacteria	1–10 μm
Mycology	Fungi	3–15 μm diameter (hyphae)
Protozoology	Protozoa	1–150 μm
Helminthology	Nematodes	3 mm–30 cm
	Platyhelminthes (cestodes, trematodes)	1 mm–10 m
Entomology	Ticks and mites	0.1–15 mm
Zoology	Horsehair worms	10–20 cm
	Mesozoa	Up to 100 cm
	Leeches	1–5 cm

Table 34.2 Various Types of Infections Associated with Parasitic Organisms

Type	Definition
Abscess	A localized infection with a collection of pus surrounded by an inflamed area
Acute	Short but severe course
Bacteremia	Presence of viable bacteria in the blood
Chronic	Persists over a long time
Covert	Subclinical, no symptoms
Cross	Transmitted between hosts infected with different organisms
Focal	Exists in circumscribed areas
Fulminating	Infectious agent multiplies with great intensity
Iatrogenic	Caused as a result of health care
Latent	Persists in tissues for long periods, during most of which there are no symptoms
Localized	Restricted to a limited region or to one or more anatomical areas
Mixed	More than one organism present simultaneously
Nosocomial	Develops during a stay at a hospital or other clinical care facility
Opportunistic	Due to an agent that does not harm a healthy host but takes advantage of an unhealthy one
Overt	Symptomatic
Phylogenetic	Caused by plant pathogens
Primary	First infection that often allows other organisms to appear on the scene
Pyogenic	Results in pus formation
Secondary	Caused by an organism following an initial or primary infection
Sepsis	(1) The condition resulting from the presence of bacteria or their toxins in blood or tissues; the presence of pathogens or their toxins in the blood or other tissues (2) Systemic response to infection; this systemic response is manifested by two or more of the following conditions as a result of infection: temperature, >38 or $<36^{\circ}\text{C}$; heart rate, >90 beats per min; respiratory rate, >20 breaths per min, or pCO_2 , <32 mm Hg; leukocyte count, $>12,000$ cells per mm^3 , or $>10\%$ immature (band) forms
Septicemia	Blood poisoning associated with persistence of pathogenic organisms or their toxins in the blood
Septic shock	Sepsis with hypotension despite adequate fluid resuscitation, along with the presence of perfusion abnormalities that may include, but are not limited to, lactic acidosis, oliguria, or an acute alteration in mental status
Severe sepsis	Sepsis associated with organ dysfunction, hypoperfusion, or hypotension; hypoperfusion and perfusion abnormalities may include, but are not limited to, lactic acidosis, oliguria, or an acute alteration in mental status
Sporadic	Occurs only occasionally
Subclinical (inapparent or covert)	No detectable symptoms or manifestations
Systemic	Spread throughout the body
Toxemia	Condition arising from toxins in the blood
Zoonosis	Caused by a parasitic organism that is normally found in animals other than humans

ternally, it is an **endoparasite**. The host on or in which the parasitic organism either attains sexual maturity or reproduces is the **final host**. A host that serves as a temporary but essential environment for some stages of development is an **intermediate host**. In contrast, a **transfer host** is not necessary for the completion of the organism's life cycle but is used as a vehicle for reaching a final host. A host infected with a parasitic organism that also can infect humans is called a **reservoir host**.

Because, by definition, parasitic organisms are dependent on their hosts, the symbiotic relationship between the host and parasite is a dynamic one (**figure 34.1**). When a parasite is growing and multiplying within or on a host, the host is said to have an **infection**. The nature of an infection can vary widely with respect to severity, location, and number of organisms involved (**table 34.2**). An infection may or may not result in overt disease. An **infectious disease** is any change from a state of health in which part or all of the host body is not capable of carrying on its normal functions due to the presence of an organism or its

products. Any organism or agent that produces such a disease is a **pathogen** [Greek *patho*, disease, and *gennan*, to produce]. Its ability to cause disease is called **pathogenicity**. A **primary (frank) pathogen** is any organism that causes disease in a healthy host by a direct interaction. Conversely, an **opportunistic pathogen** is an organism that is either normally free-living, or a part of the host's normal microbiota, but which may adopt a pathogenic role under certain circumstances, such as when the immune system is compromised.

At times an infectious organism can enter a latent state in which there is no shedding of the organism and no symptoms present within the host. This latency can be either intermittent or quiescent. Intermittent latency is exemplified by the herpes virus that causes cold sores (fever blisters). After an initial infection, the symptoms subside. However, the virus remains in nerve tissue and can be activated weeks or years later by factors such as stress or sunlight. In a quiescent latency the organism persists but remains inactive for long periods of time, usually for years. For

Figure 34.1 Symbiosis. All symbiotic relationships are dynamic, and shifts among them can occur as indicated by the arrows. The most beneficial relationship is mutualism; the most destructive is parasitism. Host susceptibility, virulence of the parasitic organism, and number of parasites are factors that influence these relationships. Disease can result from a shift from either mutualism or commensalism to parasitism. Health may be regained by the reestablishment of mutualism or commensalism.

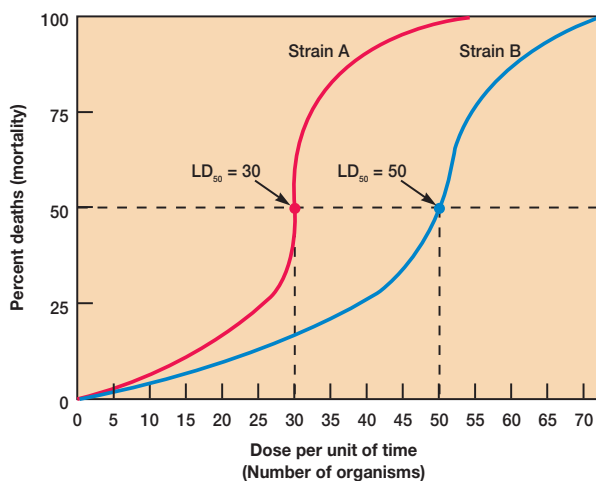
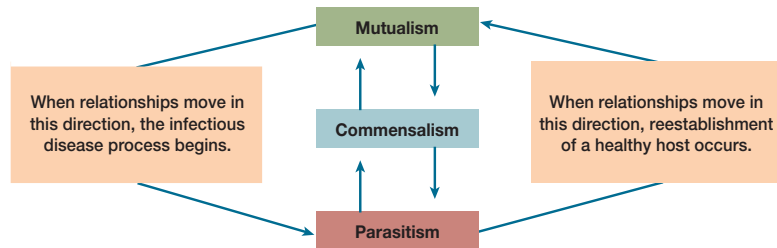


Figure 34.2 Determination of the LD₅₀ of a Pathogenic Microorganism. Various doses of a specific pathogen are injected into experimental host animals. Deaths are recorded and a graph constructed. In this example, the graph represents the susceptibility of host animals to two different strains of a pathogen—strain A and strain B. For strain A the LD₅₀ is 30, and for strain B it is 50. Hence strain A is more virulent than strain B.

example, the varicella-zoster virus causes chickenpox in children and remains after the disease has subsided. In adulthood, under certain conditions, the same virus may erupt into a disease called shingles. [Cold sores](#) (p. 884); [Chickenpox \(varicella\) and shingles \(herpes zoster\)](#) (pp. 871–72)

The outcome of most host-parasite relationships is dependent on three main factors: (1) the number of organisms present in or on the host, (2) the virulence of the organism, and (3) the host's defenses or degree of resistance. Usually the greater the number of organisms within a given host, the greater the likelihood of disease. However, a few organisms can cause disease if they are extremely virulent or if the host's resistance is low. A host's resistance can drop so much that its own microbiota may cause disease. Such a disease is sometimes called an endogenous disease because the agent originally comes from within the host's own body. Endogenous diseases can be a serious problem among hospitalized patients with very low resistance.

The term **virulence** [Latin *virulentia*, from *virus*, poison] refers to the degree or intensity of pathogenicity. It is determined

by three characteristics of the pathogen: invasiveness, infectivity, and pathogenic potential. **Invasiveness** is the ability of the organism to spread to adjacent or other tissues. **Infectivity** is the ability of the organism to establish a focal point of infection. **Pathogenic potential** refers to the degree that the pathogen causes damage. A major aspect of pathogenic potential is toxigenicity. **Toxigenicity** is the pathogen's ability to produce toxins, chemical substances that will damage the host and produce disease. Virulence is often measured experimentally by determining the **lethal dose 50 (LD₅₀)** or the **infectious dose 50 (ID₅₀)**. These values refer to the dose or number of pathogens that will either kill or infect, respectively, 50% of an experimental group of hosts within a specified period (**figure 34.2**).

It should be noted that disease can result from causes other than toxin production. Sometimes a host will trigger exaggerated immunological responses (**immunopathology**) upon a second exposure or chronic exposure to a microbial antigen. These hypersensitivity reactions damage the host even though the pathogen doesn't produce a toxin. Tuberculosis is a good example of the involvement of hypersensitivity reactions in disease (*see pp. 906–8*). Some diseases also might be due to autoimmune responses. For instance, a viral or bacterial pathogen may stimulate the immune system to attack host tissues because it carries antigens that resembled those of the host, a phenomenon known as molecular mimicry. Streptococcal infections may cause rheumatic fever in this way (*see p. 905*). [Hypersensitivity reactions](#) (pp. 768–71)

1. Define parasitic organism, parasitism, infection, infectious disease, pathogenicity, virulence, invasiveness, infectivity, pathogenic potential, and toxigenicity.
2. What factors determine the outcome of most host-parasite relationships?

34.2 Pathogenesis of Viral Diseases

The fundamental process of viral infection is the expression of the viral replicative cycle (*see figures 17.5, 18.4–18.7*) in a host cell. The steps for the infectious process involving viruses are that a virus must

1. Enter a host
2. Contact and enter susceptible cells

3. Replicate within the cells
4. Spread to adjacent cells
5. Cause cellular injury
6. Engender a host immune response
7. Be either cleared from the body of the host, establish a persistent infection, or kill the host
8. Be shed back into the environment

The determinants of pathogenicity are now discussed in more detail.

Entry, Contact, and Primary Replication

The first step in the infectious process is the attachment and entrance of the virus into a susceptible host and the host's cells.

Entrance may be accomplished through one of the body surfaces (skin, respiratory system, gastrointestinal system, urogenital system, or the conjunctiva of the eye). Other viruses enter the host by needle sticks, blood transfusions and organ transplants, or by insect **vectors** (organisms that transmit the pathogen from one host to another). Some viruses replicate at the site of entry, cause disease at the same site (e.g., respiratory and gastrointestinal infections), and do not spread throughout the body. Others spread to sites distant from the point of entry and replicate at these sites. For example, the enteroviruses enter through the gastrointestinal tract but produce disease in the central nervous system. [Mucous membranes \(pp. 710–12\)](#)

Viral Spread and Cell Tropism

Mechanisms of viral spread vary, but the most common routes are the bloodstream and lymphatic system. The presence of viruses in the blood is called **viremia**. In some instances, spread is by way of nerves (e.g., rabies virus, herpes simplex, and varicella-zoster viruses; *see figure 38.2*).

Viruses exhibit cell, tissue, and organ specificities. These specificities are called **tropisms** (Greek *trope*, turning). A tropism by a specific virus usually reflects the presence of specific cell surface receptors on the eucaryotic host cell for that virus (*see figures 18.4 and 38.14*).

Cell Injury and Clinical Illness

Destruction of the virus-infected cells in the target tissues and alterations in host physiology are responsible for the development of viral disease and clinical illness. Some tissue (e.g., intestinal epithelium) can rapidly regenerate after a viral attack and withstand extensive damage. Other tissues, such as nervous system tissues, are not able to regenerate and may never resume normal functioning after damage has occurred.

The potential effects of viruses on individual host cells are the result of a complex series of events. There are four generally accepted patterns of a viral infection. (1) In lytic infections the virus multiplies and kills the host cell immediately and new virions are released. (2) In persistent viral infections the virus lives in the host cell and releases small numbers of virions over a long period of time. This causes little damage to the host cell. (3) In latent infections, the virus resides in the cell but produces no viri-

ons. At some later time the virus can be activated and a lytic infection occurs (*see section 18.4*). (4) Some viruses can transform the host cell into a cancer cell that becomes the focal point for a tumor (*see section 18.5*).

Host Immune Response

Both humoral and cellular components of the immune response are involved in the control of viral infections and were discussed in detail in chapters 31 and 32.

Recovery from Infection

The host will either succumb or recover from a viral infection. Recovery mechanisms involve nonspecific defense mechanisms and specific humoral and cellular immunity. The relative importance of each of these factors varies with the virus and the disease, and will be covered in chapter 38.

Virus Shedding

The last step in the infectious process is shedding of the infectious virus back into the environment. This is necessary to maintain a source of viruses in a population of hosts. Shedding often occurs from the same body surface used for entry. During this period, an infected host is infectious and can spread the virus. In some viral infections, such as a rabies infection, humans are dead-end hosts because virus shedding does not occur.

1. For a virus to cause disease, certain steps must be accomplished. Briefly describe each of these steps.
2. What are the four most common patterns of viral infections?

34.3 Pathogenesis of Bacterial Diseases

The steps for infections by pathogenic bacteria include

1. Maintain a reservoir. A **reservoir** is a place to live before and after causing an infection (*see p. 854*).
2. Initially be transported to the host.
3. Adhere to, colonize, and/or invade the host.
4. Multiply (grow) or complete its life cycle on or in the host or the host's cells.
5. Initially evade host defense mechanisms.
6. Possess the ability to damage the host.
7. Leave the host and return to the reservoir or enter a new host.

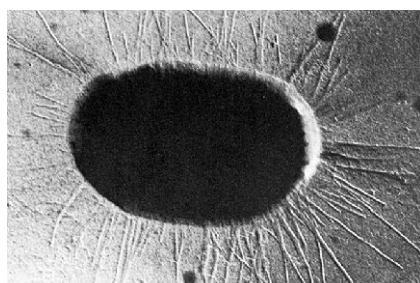
The first five factors influence the degree of infectivity and invasiveness. Toxigenicity plays a major role in the sixth. The determinants are now discussed in more detail.

Maintaining a Reservoir of the Bacterial Pathogen

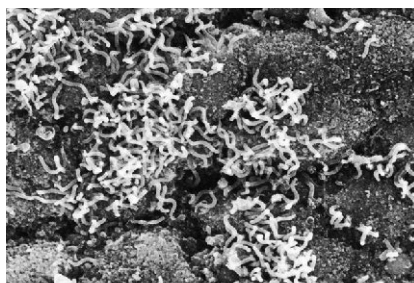
All bacterial pathogens must have at least one reservoir. The most common reservoirs for human pathogens are other humans, animals, and the environment. Since the source and/or reservoir of

Table 34.3 Bacterial Adherence Factors (Adhesins) That Play a Role in Infectious Diseases

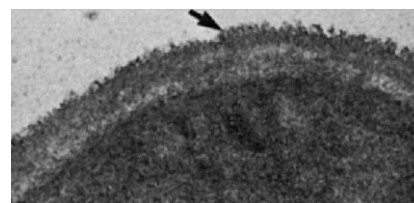
Adherence Factor	Description
Fimbriae	Filamentous structures that help attach bacteria to other bacteria or to solid surfaces
Glycocalyx or capsule	A layer of exopolysaccharide fibers with a distinct outer margin that surrounds many cells; it inhibits phagocytosis and aids in adherence; when the layer is well organized and not easily washed off it is called a capsule
Pili	Filamentous structures that bind procaryotes together for the transfer of genetic material
S layer	The outermost regularly structured layer of cell envelopes of some archaeobacteria and eubacteria that may promote adherence to surfaces
Slime layer	A bacterial film that is less compact than a capsule and is removed easily
Teichoic and lipoteichoic acids	Cell wall components in gram-positive bacteria that aid in adhesion



(a)



(b)



(c)

Figure 34.3 Microbial Adherence. (a) Transmission electron micrograph of fimbriated *Escherichia coli* ($\times 16,625$). (b) Scanning electron micrograph of epithelial cells with adhering vibrios ($\times 1,200$). (c) *Candida albicans* fimbriae (arrow) are used to attach the fungus to vaginal epithelial cells.

the pathogen is part of the infectious disease cycle, this aspect of pathogenicity is discussed in detail in chapter 37, which covers the epidemiology of infectious diseases. [The infectious disease cycle](#) (pp. 852–58)

Transport of the Bacterial Pathogen to the Host

An essential feature in the development of an infectious disease is the initial transport of the bacterial pathogen to the host. The most obvious means is direct contact—from host to host (coughing, sneezing, body contact). Bacteria also are transmitted indirectly in a variety of ways. Infected hosts shed bacteria into their surroundings. Once in the environment bacteria can be deposited on various surfaces, from which they can be either resuspended into the air or indirectly transmitted to a host later. Soil, water, and food are indirect vehicles that harbor and transmit bacteria to hosts. Vectors and **fomites** (inanimate objects that harbor and transmit pathogens) also are involved in the spread of many bacteria.

Attachment and Colonization by the Bacterial Pathogen

After being transmitted to an appropriate host, the bacterial pathogen must be able to adhere to and colonize host cells and tissues. In this context **colonization** means the establishment of a site of microbial reproduction on or within a host. It does not necessarily result in tissue invasion or damage. Colonization depends on the ability of the bacteria to compete successfully with the

host's normal microbiota for essential nutrients. Specialized structures that allow bacteria to compete for surface attachment sites also are necessary for colonization.

Bacterial pathogens and many nonpathogens adhere with a high degree of specificity to particular tissues. Adherence factors called **adhesins** (table 34.3) are one reason for this specificity. Adhesins are specialized molecules or structures on the bacteria's cell surface that bind to complementary receptor sites on the host cell surface (figure 34.3). They are one type of virulence factor. **Virulence factors** are bacterial products or structural components (e.g., capsules and adhesins) that contribute to virulence or pathogenicity.

Invasion of the Bacterial Pathogen

Entry into host cells and tissues is a specialized strategy used by many bacterial pathogens for survival and multiplication. Pathogens often actively penetrate the host's mucous membranes and epithelium after attachment to the epithelial surface. This may be accomplished through production of lytic substances that alter the host tissue by (1) attacking the ground substance and basement membranes of integuments and intestinal linings, (2) degrading carbohydrate-protein complexes between cells or on the cell surface (the glycocalyx), or (3) disrupting the cell surface.

At times a bacterial pathogen can penetrate the epithelial surface by passive mechanisms not related to the pathogen itself. Examples include (1) small breaks, lesions, or ulcers in a mu-

Table 34.4 Microbial Products (Virulence Factors) Involved in Bacterial Pathogen Dissemination Throughout a Mammalian Host

Product	Organism Involved	Mechanism of Action
Coagulase	<i>Staphylococcus aureus</i>	Coagulates (clots) the fibrinogen in plasma. The clot protects the pathogen from phagocytosis and isolates it from other host defenses.
Collagenase	<i>Clostridium</i> spp.	Breaks down collagen that forms the framework of connective tissues; allows the pathogen to spread.
Deoxyribonuclease (along with calcium and magnesium)	Group A streptococci, staphylococci, <i>Clostridium perfringens</i>	Lowers viscosity of exudates, giving the pathogen more mobility.
Elastase and alkaline protease	<i>Pseudomonas aeruginosa</i>	Cleaves laminin associated with basement membranes.
Hemolysins	Staphylococci, streptococci, <i>Escherichia coli</i> , <i>Clostridium perfringens</i>	Lyse erythrocytes, causing anemia and weakened host defenses; make iron available for microbial growth.
Hyaluronidase	Groups A, B, C, and G streptococci, staphylococci, clostridia	Hydrolyzes hyaluronic acid, a constituent of the intercellular ground substance that cements cells together and renders the intercellular spaces amenable to passage by the pathogen.
Hydrogen peroxide (H ₂ O ₂) and ammonia (NH ₃)	<i>Mycoplasma</i> spp., <i>Ureaplasma</i> spp.	Are produced as metabolic wastes. These are toxic and damage epithelia in respiratory and urogenital systems.
Immunoglobulin A protease	<i>Streptococcus pneumoniae</i>	Cleaves immunoglobulin A into Fab and Fc fragments.
Lecithinase or phospholipase	<i>Clostridium</i> spp.	Destroys the lecithin (phosphatidylcholine) component of plasma membranes, allowing pathogen to spread.
Leukocidins	Staphylococci, pneumococci, streptococci	Pore-forming exotoxins that kill leukocytes; cause degranulation of lysosomes within leukocytes, which decreases host resistance.
Porins	<i>Salmonella typhimurium</i>	Inhibit leukocyte phagocytosis by activating the adenylate cyclase system.
Protein A	<i>Staphylococcus aureus</i>	Located on cell wall. Immunoglobulin G (IgG) binds to protein A by its Fc end, thereby preventing complement from interacting with bound IgG.
Pyrogenic Exotoxin B (cysteine protease)	Group A streptococci, <i>Streptococcus pyogenes</i>	Degrades proteins.
Streptokinase (fibrinolysin, staphylokinase)	Group A, C, and G streptococci, staphylococci	A protein that binds to plasminogen and activates the production of plasmin, thus digesting fibrin clots; this allows the pathogen to move from the clotted area.

cous membrane that permit initial entry; (2) wounds, abrasions, or burns on the skin's surface; (3) arthropod vectors that create small wounds while feeding; (4) tissue damage caused by other organisms; and (5) existing eucaryotic internalization pathways (e.g., endocytosis and phagocytoses; see figure 31.16).

Once under the mucous membrane, the bacterial pathogen may penetrate to deeper tissues and continue disseminating throughout the body of the host. One way the pathogen accomplishes this is by producing specific products and/or enzymes that promote spreading (table 34.4). These products are virulence factors. Bacteria may also enter the small terminal lymphatic capillaries that surround epithelial cells. These capillaries merge into large lymphatic vessels that eventually drain into the circulatory system. Once the circulatory system is reached, the bacteria have access to all organs and systems of the host.

Growth and Multiplication of the Bacterial Pathogen

For a bacterial pathogen to be successful in growth and reproduction (colonization), it must find an appropriate environment (nutrients, pH, temperature, redox potential) within the host. Those areas of the host's body that provide the most favorable conditions will harbor the pathogen and allow it to grow and multiply to produce an infection. Some bacteria invade specific cells in which they grow and

multiply. Many of these intracellular pathogens have evolved such elaborate nutrient-gathering mechanisms that they have become totally dependent on the host's cells. Finally, some bacteria can actively grow and multiply in the blood plasma. The presence of viable bacteria in the bloodstream is called **bacteremia**. The presence of bacteria or their toxins in the blood often is termed **septicemia** [Greek *septikos*, produced by putrefaction, and *haima*, blood].

Leaving the Host

The last determinant of a successful bacterial pathogen is its ability to leave the host and enter either a new host or a reservoir. Unless a successful escape occurs, the disease cycle will be interrupted and the microorganism will not be perpetuated. Most bacteria employ passive escape mechanisms. Passive escape occurs when a pathogen or its progeny leave the host in feces, urine, droplets, saliva, or desquamated cells.

The Clonal Nature of Bacterial Pathogens

The main mechanism bacteria use to exchange genetic information is the transfer of extrachromosomal genetic elements, plasmids, and phages (see chapter 13). Many genes that code for bacterial virulence factors are found on plasmids or phages. These mobile genetic elements can transfer virulence factors between members

of the same species or different species by horizontal gene transfer (see section 13.1). At times the genetic elements are part of highly mobile DNA (transposons, see section 13.3), and there is a recombination between the extrachromosomal DNA and the chromosome. When this recombination occurs, the genes coding for virulence may become chromosomal.

One important result of the conservation of these chromosomal genes is that the bacteria are clonal. Some bacterial pathogens have only one or a few clonal types existing in the environment. For example, the bacterium that causes typhoid fever (*Salmonella typhi*) has two clonal types, whereas there are over 100 clonal types of *Haemophilus influenzae*, but only a small number are associated with bacterial pneumonia.

Regulation of Bacterial Virulence Factors

As noted in many chapters, some pathogenic bacteria have adapted to both the free-living state and to an environment within a human host. In the adaptive process, these pathogens have evolved complex signal transduction pathways to regulate the genes necessary for virulence. A virulence factor may be present simply because the bacterium has been infected by a phage. Often environmental factors control the expression of the virulence genes. Common signals include temperature, osmolality, available iron, pH, specific ions, and other nutrient factors. Several examples are now presented.

The gene for diphtheria toxin (figure 34.5b) from *Corynebacterium diphtheriae* (the pathogen that causes diphtheria) is carried on the temperate bacteriophage β , and its expression is regulated by iron. The toxin is produced only by strains lysogenized by the phages. Expression of the virulence gene of *Bordetella pertussis* (the pathogen that causes whooping cough) is enhanced when the bacteria grow at body temperature (37°C) and suppressed when grown at a lower temperature. Finally, the virulence factors of *Vibrio cholerae* (the pathogen that causes cholera) are regulated at various levels by many environmental factors. Expression of the cholera toxin is higher at pH 6 than at pH 8 and higher at 30 than at 37°C. Osmolality and available amino acids are also important.

Pathogenicity Islands

Many bacteria (*Yersinia* spp., *Pseudomonas aeruginosa*, *Shigella flexneri*, *Salmonella typhimurium*, enteropathogenic *E. coli*) are pathogenic because they have large segments of DNA, called **pathogenicity islands**, that carry genes responsible for virulence. These pathogenicity islands have been acquired during evolution by horizontal gene transfer. A pathogen may have more than one pathogenicity island. An excellent example of virulence genes carried in a pathogenicity island are those involved in protein secretion. So far, five pathways of protein secretion (types I to V) have been described in gram-negative bacteria. A set of approximately 20 genes encode a pathogenicity mechanism termed the **type III secretion system** that enables gram-negative bacteria to secrete and inject virulence proteins into the cytoplasm of eucaryotic host cells (figure 34.4). Unlike other bacterial secretory systems, the type III system is triggered specifically by contact with host cells, which helps avoid inappropriate activation of host defenses. Secretion of these virulence proteins into a host cell initiates sophisticated “biochemical cross-talk” between the pathogen and the host. The injected proteins re-

semble eucaryotic factors that signal transduction functions and they are capable of interfering with eucaryotic signaling pathways. Redirection of cellular signaling transduction may disarm host immune responses or reorganize the cytoskeleton, thus establishing subcellular niches for bacterial colonization and facilitating “stealth and interdiction” of host defense communication lines.

Pathogenicity islands generally increase microbial virulence and are not present in nonpathogenic members of the same genus or species. One specific example is found in *E. coli*. The enteropathogenic *E. coli* possesses large DNA fragments, 35 to 170 kilobases in size, that contain several virulence genes absent from commensal *E. coli*. Some of these genes code for proteins that alter actin microfilaments within a host intestinal cell. As a consequence, the host cell surface bulges and develops into a cuplike pedestal to which the bacterium tightly binds.

1. What seven steps are involved in the infection process and pathogenesis of bacterial diseases?
2. What are some ways in which bacterial pathogens are transmitted to their hosts? Define vector and fomite.
3. Describe several specific adhesins by which bacterial pathogens attach to host cells.
4. Once under the mucous and epithelial surfaces, what are some mechanisms that bacterial pathogens possess to promote their dissemination throughout the body of a host?
5. What are virulence factors? Pathogenicity islands?
6. What is the significance of the clonal nature of bacterial pathogens? The regulation of virulence factors?

Toxigenicity

Two distinct categories of disease can be recognized based on the role of the bacteria in the disease-causing process: infections and intoxications. An infectious disease results partly from the pathogen’s growth and reproduction (or invasiveness) that often produce tissue alterations.

Intoxications are diseases that result from the entrance of a specific preformed toxin (e.g., botulinum toxin) into the body of a host. Toxins can even induce disease in the absence of the organism that produced them. A **toxin** [Latin *toxicum*, poison] is a substance, such as a metabolic product of the organism, that alters the normal metabolism of host cells with deleterious effects on the host. The term **toxemia** refers to the condition caused by toxins that have entered the blood of the host. Toxins produced by bacteria can be divided into two main categories: exotoxins and endotoxins. The primary characteristics of the two groups are compared in table 34.5.

Exotoxins

Exotoxins are soluble, heat-labile, proteins (a few are enzymes) that usually are released into the surroundings as the bacterial pathogen grows. Often exotoxins may travel from the site of infection to other body tissues or target cells in which they exert their effects. Exotoxins usually are

1. Synthesized by specific bacteria that often have plasmids or prophages bearing the exotoxin genes

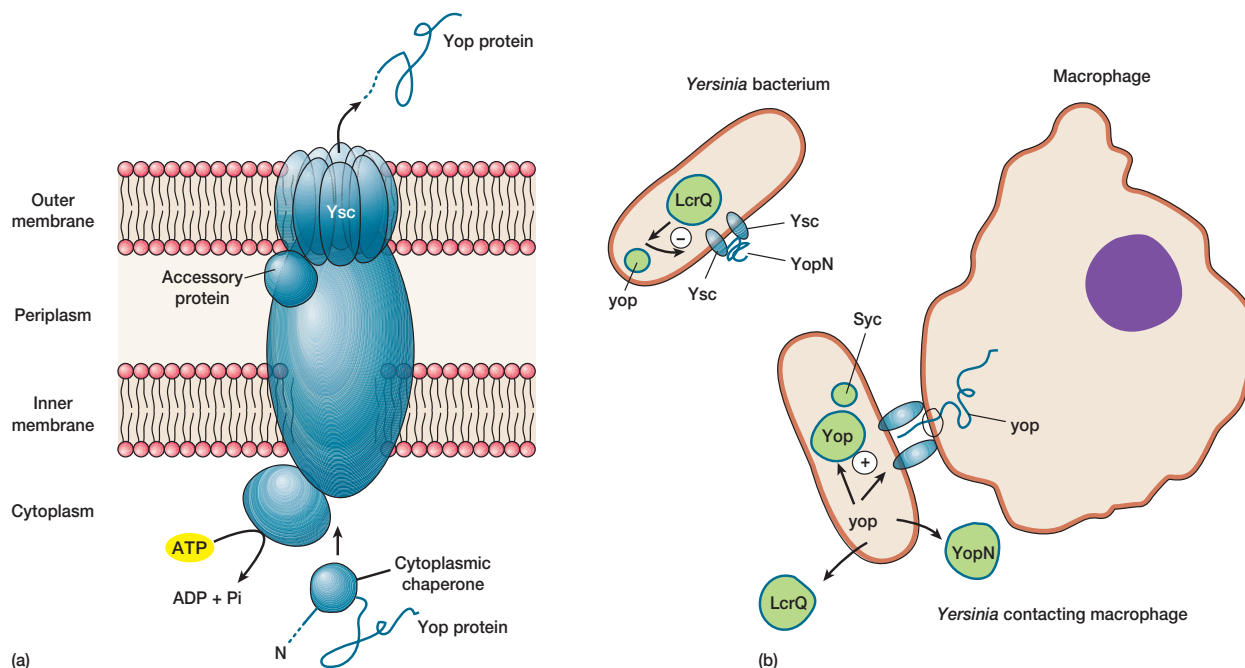


Figure 34.4 Type III Secretion System and Mode of Action of Yop Virulence Protein. (a) Schematic drawing of the type III secretion system and Yop (a specific virulence protein) secretion by *Yersinia* spp. (b) In this schematic, prior to cell contact, the type III secretion channels (Ysc) are kept shut by YopN, and cytoplasmic accumulation of the negative regulatory factor LcrQ leads to transcription repression of *yop* genes (negative sign in circle). After *Yersinia* contacts the macrophage, YopN is released, allowing rapid secretion of LcrQ, which in turn relieves the block of *yop* gene expression (positive sign in circle). Anti-host Yop proteins, protected by cognate chaperones (Syc), are translocated into the macrophage by the type III secretion machinery (as shown in part a). After contact of the *Yersinia* bacterium with a macrophage, Yop is injected into the cytoplasm of the target cell where it catalyzes a rapid and specific dephosphorylation of several macrophage proteins that are required for normal phagocytosis.

Table 34.5 Characteristics of Exotoxins and Endotoxins

Characteristic	Exotoxins	Endotoxins
Chemical composition	Protein, often with two components (A and B)	Lipopolysaccharide complex on outer membrane; lipid A portion is toxic
Disease examples	Botulism, diphtheria, tetanus	Gram-negative infections, meningococcemia
Effect on host	Highly variable between different toxins	Similar for all endotoxins
Fever	Usually do not produce fever	Produce fever by release of interleukin-1
Genetics	Frequently carried by extrachromosomal genes such as plasmids	Synthesized directly by chromosomal genes
Heat stability	Most are heat sensitive and inactivated at 60–80°C	Heat stable
Immune response	Antitoxins provide host immunity; highly antigenic	Limited antibodies produced; weakly immunogenic
Location	Usually excreted outside the living cell	Part of outer membrane of gram-negative bacteria
Production	Produced by both gram-positive and gram-negative bacteria	Found only in gram-negative bacteria; Released on bacterial death and some liberated during growth.
Toxicity	Highly toxic and fatal in microgram quantities	Moderate toxicity
Toxoid production	Converted to antigenic, nontoxic toxoids; toxoids are used to immunize (e.g., tetanus toxoid)	Toxoids cannot be made

- Heat-labile proteins inactivated at 60 to 80°C
- Among the most lethal substances known (toxic in very small doses [microgram per kilogram amounts]; e.g., the botulinum toxin)
- Associated with specific diseases and have specific mechanisms of action
- Highly immunogenic and stimulate the production of neutralizing antibodies called **antitoxins**
- Easily inactivated by formaldehyde, iodine, and other chemicals to form immunogenic **toxoids**
- Unable to produce a fever in the host directly
- Often given the name of the disease they produce (e.g., the diphtheria toxin)

Exotoxins can be divided into four types based on their structure and physiological activities. (1) One type is the AB toxin, which gets its name from the fact that the portion of the toxin (B) that binds to a host cell receptor is separate from the portion (A) that has the enzyme activity that causes the toxicity. (2) A second type, which also may be an AB toxin, consists of those toxins that affect a specific host site (nervous tissue [neurotoxins], the intestines [enterotoxins], general tissues [cytotoxins]) by acting extracellularly or intracellularly on the host cells. (3) A third type does not have separable A and B portions and acts by disorganizing host cell membranes. Examples include the leukocidins, hemolysins, and phospholipases. (4) A fourth type is the superantigen that acts by stimulating T cells to release cytokines. Examples of the first three types are now discussed. Superantigens are discussed in detail in section 32.2. The general properties of some AB exotoxins are presented in **table 34.6**.

Table 34.6 Properties of Some AB Model Bacterial Exotoxins

Toxin	Organism	Genetic Control	Subunit Structure	Target Cell Receptor	Enzymatic Activity	Biologic Effects
Anthrax toxins	<i>B. anthracis</i>	Plasmid	Three separate proteins (EF, LF, PA) ^a	Unknown, probably glycoprotein	EF is a calmodulin-dependent adenylate cyclase; LF enzyme activity is unknown	EF + PA: increase in target cell cAMP level, localized edema; LF + PA: death of target cells and experimental animals
<i>Bordetella</i> adenylate cyclase toxin	<i>Bordetella</i> spp.	Chromosomal	A-B ^b	Unknown, probably glycolipid	Calmodulin-activated cyclase	Increase in target cell cAMP level; modified cell function or cell death
Botulinum toxin	<i>C. botulinum</i>	Phage	A-B ^c	Possibly ganglioside (GD _{1b})	Zinc-dependent endopeptidase cleavage of synaptobrevin	Decrease in peripheral, presynaptic acetylcholine release; flaccid paralysis
Cholera toxin	<i>V. cholera</i>	Chromosomal	A-5B ^d	Ganglioside (GM ₁)	ADP ribosylation of adenylate cyclase regulatory protein, G _s	Activation of adenylate cyclase, increase in cAMP level; secretory diarrhea
Diphtheria toxin	<i>C. diphtheriae</i>	Phage	A-B ^e	Heparin-binding, EGF-like growth factor precursor	ADP ribosylation of elongation factor 2	Inhibition of protein synthesis; cell death
Heat-labile enterotoxins ^f	<i>E. coli</i>	Plasmid	—————	Similar or Identical to Cholera Toxin	—————	—————
Pertussis toxin	<i>B. pertussis</i>	Chromosomal	A-5B ^g	Unknown, probably glycoprotein	ADP ribosylation of signal-transducing G proteins	Block of signal transduction mediated by target G proteins
<i>Pseudomonas</i> exotoxin A	<i>P. aeruginosa</i>	Chromosomal	A-B	α ₂ -Macroglobulin/LDL receptor	————— Similar or Identical to Diphtheria Toxin	—————
Shiga toxin	<i>S. dysenteriae</i>	Chromosomal	A-5B ^h	Globotriaosylceramide (Gb ₃)	RNA N-glycosidase	Inhibition of protein synthesis, cell death
Shiga-like toxin 1	<i>Shigella</i> spp., <i>E. coli</i>	Phage	—————	Similar or Identical to Shiga Toxin	—————	—————
Tetanus toxin	<i>C. tetani</i>	Plasmid	A-B ^c	Ganglioside (GT ₁ and/or GD _{1b})	Zinc-dependent endopeptidase cleavage of synaptobrevin	Decrease in neurotransmitter release from inhibitory neurons; spastic paralysis

Adapted from G. L. Mandell, et al., *Principles and Practice of Infectious Diseases*, 3d edition Copyright © 1990 Churchill-Livingstone, Inc., Medical Publishers, New York, NY. Reprinted by permission.

^aThe binding component (known as protective antigen [PA]) catalyzes/facilitates the entry of either edema factor (EF) or lethal factor (LF).

^bApparently synthesized as a single polypeptide with binding and catalytic (adenylate cyclase) domains.

^cHolotoxin is apparently synthesized as a single polypeptide and cleaved proteolytically as diphtheria toxin; subunits are referred to as L: light chain, A equivalent; H: heavy chain, B equivalent.

^dThe A subunit is proteolytically cleaved into A₁ and A₂, with A₁ possessing the ADP-ribosyl transferase activity; the binding component is made up of five identical B units.

^eHolotoxin is synthesized as a single polypeptide and cleaved proteolytically into A and B components held together by disulfide bonds.

^fThe heat-labile enterotoxins of *E. coli* are now recognized to be a family of related molecules with identical mechanisms of action.

^gThe binding portion is made up of two dissimilar heterodimers labeled S2-S3 and S2-S4 that are held together by a bridging peptide, SS.

^hSubunit composition and structure similar to cholera toxin.

AB Exotoxins. **AB toxins** are composed of an enzymatic subunit or fragment (A) that is responsible for the toxic effect once inside the host cell and a binding subunit or fragment (B). Isolated A subunits are enzymatically active but lack binding and cell entry capability, whereas isolated B subunits bind to target cells but are nontoxic and biologically inactive. The B subunit interacts with specific receptors on the target cell or tissue such as the gangliosides GM₁ for cholera toxin, GT₁ and/or GD₁ for tetanus toxin, and GD₁ for botulinum toxin.

Several mechanisms for the entry of A subunits or fragments into target cells have been proposed. In one mechanism the B subunit inserts into the plasma membrane and creates a pore through which the A subunit enters (**figure 34.5a**). In another mechanism entry is by receptor-mediated endocytosis (**figure 34.5b**).

The mechanism of action of an AB toxin can be quite complex, as shown by the example of diphtheria toxin (**figure 34.5b**). The diphtheria toxin is a protein of about 62,000 mol wt. It binds to cell surface receptors by the B fragment portion and is taken into the cell through the formation of a clathrin-coated vesicle (*see p. 403*). The toxin then enters the vesicle membrane and is cleaved into two parts, one of which, the A fragment, escapes into the cytosol. The A fragment is an enzyme that catalyzes the addition of an ADP-ribose group to the eucaryotic elongation factor EF2 that aids in translocation during protein synthesis (*see pp. 270–71*). The substrate for this reaction is the coenzyme NAD⁺.



The modified EF2 protein cannot participate in the elongation cycle of protein synthesis, and the cell dies because it can no longer synthesize proteins.

AB exotoxins vary widely in their relative contribution to the disease process with which they are associated.

Specific Host Site Exotoxins. The second type of exotoxin is categorized on the basis of the site affected: **neurotoxins** (nerve tissue), **enterotoxins** (intestinal mucosa), and **cytotoxins** (general tissues). Some of the bacterial pathogens that produce these exotoxins are presented in table 34.6: neurotoxins (botulinum toxin and tetanus toxin), enterotoxins (cholera toxin, *E. coli* heat-labile toxins), and cytotoxins (diphtheria toxin, Shiga toxin).

Neurotoxins usually are ingested as preformed toxins that affect the nervous system and indirectly cause enteric (pertaining to the small intestine) symptoms. Examples include staphylococcal enterotoxin B, *Bacillus cereus* emetic toxin [Greek *emetos*, vomiting], and botulinum toxin.

True enterotoxins [Greek *enter*, intestine] have a direct effect on the intestinal mucosa and elicit profuse fluid secretion. The classic enterotoxin, cholera toxin (cholera toxin), has been studied extensively. It is an AB toxin. The B subunit is made of five parts arranged as a donut-shaped ring. The B subunit ring anchors itself to the epithelial cell's plasma membrane and then inserts the smaller A subunit into the cell. The A subunit activates tissue adenylate cyclase to increase intestinal cyclic AMP (cAMP) concentrations. High concentrations of cAMP provoke the movement of massive quantities of water and electrolytes across the in-

testinal cells into the lumen of the gut. The genes for this enterotoxigenicity reside on the *Vibrio cholera* chromosome. **Toxins in food** (pp. 926–33, 975–76); **Toxins as superantigens** (p. 732)

Cytotoxins have a specific toxic action upon cells/tissues of special organs and are named according to the type of cell/tissue or organ for which they are specific. Examples include nephrotoxin (kidney), hepatotoxin (liver), and cardiotoxin (heart).

Membrane-Disrupting Exotoxins. The third type of exotoxin lyses host cells by disrupting the integrity of the plasma membrane. There are two subtypes of **membrane-disrupting exotoxins**. The first (**figure 34.6a**), is a protein that binds to the cholesterol portion of the host cell plasma membrane, inserts itself into the membrane, and forms a channel (pore). This causes the cytoplasmic contents to leak out. Also, because the osmolality of the cytoplasm is higher than the extracellular fluid, this causes a sudden influx of water into the cell, causing it to swell and rupture. Two specific examples of this type of membrane-disrupting exotoxin are now presented.

Some pathogens produce membrane-disrupting toxins that kill phagocytic leukocytes. These are termed **leukocidins** [*leukocyte* and Latin *caedere*, to kill]. Most leukocidins are produced by pneumonococci, streptococci, and staphylococci. Since the pore-forming exotoxin produced by these bacteria destroys leukocytes, this in turn decreases host resistance. Other toxins, called **hemolysins** [*haima*, blood, and Greek *lysis*, dissolution], also can be secreted by pathogenic bacteria. Many hemolysins probably form pores in the plasma membrane of erythrocytes through which hemoglobin and/or ions are released (the erythrocytes lyse or, more specifically, hemolyze). **Streptolysin-O (SLO)** is a hemolysin, produced by *Streptococcus pyogenes*, that is inactivated by O₂ (hence the “O” in its name). SLO causes beta hemolysis of erythrocytes on agar plates incubated anaerobically (*see figure 23.17*). A complete zone of clearing around the bacterial colony growing on blood agar is called **beta hemolysis**, and a partial clearing of the blood is called **alpha hemolysis**. **Streptolysin-S (SLS)** is also produced by *S. pyogenes* but is insoluble and bound to the bacterial cell. It is O₂ stable (hence the “S” in its name) and causes beta hemolysis on aerobically incubated blood-agar plates. SLO and SLS act as leukocidins and kill leukocytes. It should also be noted that hemolysins attack the plasma membranes of many cells, not just erythrocytes and leukocytes.

The second sub-type of membrane-disrupting toxins are the **phospholipase** enzymes. Phospholipases remove the charged head group (**figure 34.6b**) from the lipid portion of the phospholipids in the host-cell plasma membrane. This destabilizes the membrane and the cell lyses and dies. One example of the pathogenesis caused by phospholipases is the disease gas gangrene. In this disease, the *Clostridium perfringens* α -toxin almost totally destroys the local population of white blood cells.

Roles of Exotoxins in Disease. Bacterial exotoxins affect a human host three main ways: (1) ingestion of preformed exotoxin, (2) colonization of a mucosal surface followed by exotoxin production, and (3) colonization of a wound or abscess followed by local exotoxin production. Each of these is now briefly discussed.

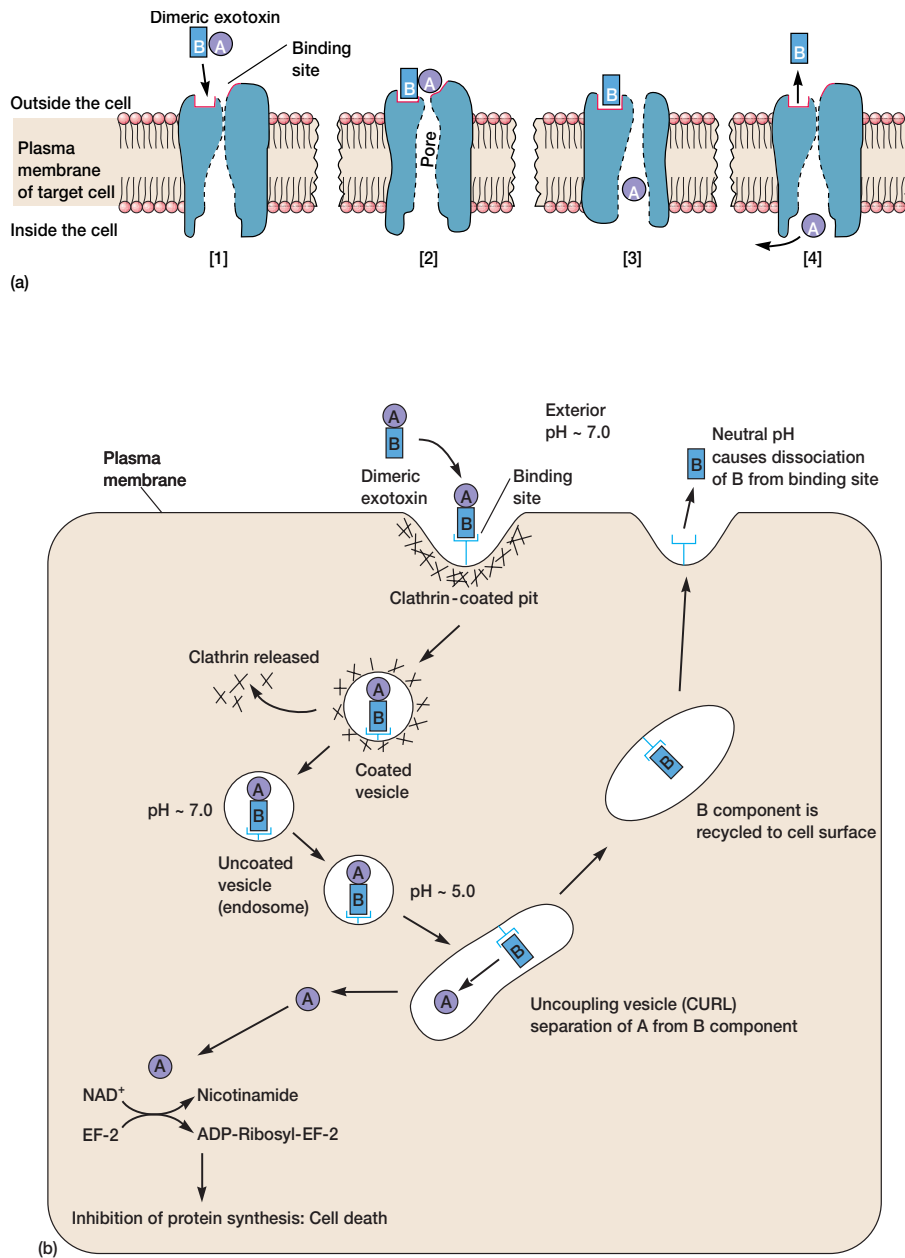


Figure 34.5 Diagrammatic Representation of Two AB Exotoxin Transport Mechanisms. (a) Domain B of the dimeric exotoxin (AB) binds to a specific membrane receptor of a target cell [1]. A conformational change [2] generates a pore [3] through which the A domain crosses the membrane and enters the cytosol, followed by recreation [4] of the binding site. (b) Receptor-mediated endocytosis of the diphtheria toxin involves the dimeric exotoxin binding to a receptor-ligand complex that is internalized in a clathrin-coated pit that pinches off to become a coated vesicle. The clathrin coat depolymerizes resulting in an uncoated endosome vesicle. The pH in the endosome decreases due to the H⁺-ATPase activity. The low pH causes A and B components to separate. An endosome in which this separation occurs is sometimes called a CURL (compartment of uncoupling of receptor and ligand). The B domain is then recycled to the cell surface. The A domain moves through the cytosol, catalyzes the ADP-ribosylation of EF-2 (elongation factor 2) and inhibits protein synthesis, leading to cell death.

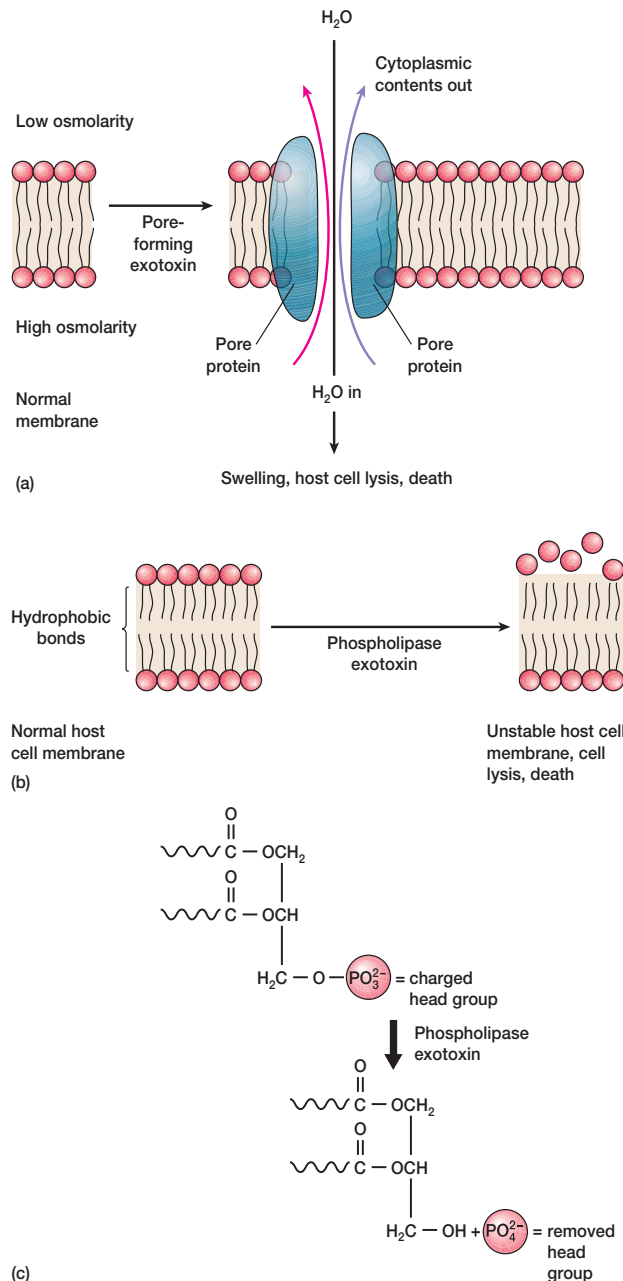


Figure 34.6 Two Subtypes of Membrane-Disrupting Exotoxins.

(a) A channel-forming (pore-forming) type of exotoxin that inserts itself into the normal host cell membrane and makes an open channel (pore). Formation of multiple pores causes cytoplasmic contents to leave the cell and water to move in, leading to cellular lysis and death of the host cell. (b) A phospholipid-hydrolyzing phospholipase exotoxin destroys membrane integrity. (c) The exotoxin removes the charged polar head groups from the phospholipid part of the host cell membrane. This destabilizes the membrane and causes the host cell to lyse.

In the first example (**figure 34.7a**), the exotoxin is produced by bacteria growing in food. When food is consumed, the preformed exotoxin is also consumed. The classical example is staphylococcal food poisoning (*see section 39.4*) caused solely by the ingestion of preformed enterotoxin. Since the bacteria (*Staphylococcus aureus*) cannot colonize the gut, they pass through the body without producing any more exotoxin; thus, this type of bacterial disease is self-limiting.

In the second example (**figure 34.7b**), bacteria colonize a mucosal surface but do not invade underlying tissue or enter the bloodstream. The toxin either causes disease locally or enters the bloodstream and is distributed systemically where it can cause disease at distant sites. The classical example here is the disease cholera caused by *Vibrio cholerae* (*see section 39.4*). Once the bacteria enter the body, they adhere to the intestinal mucosa where they are not invasive but secrete the cholera toxin, which is an AB exotoxin that catalyzes an ADP-ribosylation similar to that of diphtheria exotoxin (**figure 34.5b**). As a result, cholera toxin stimulates hypersecretion of water and chloride ions and the patient loses massive quantities of water.

The third example of exotoxins in disease pathogenesis occurs when bacteria grow in a wound or abscess (**figure 34.7c**). The exotoxin causes local tissue damage or kills phagocytes that enter the infected area. A disease of this type is gas gangrene (*see section 39.3*) in which the exotoxin (α -toxin) of *Clostridium perfringens* causes the tissue destruction in the wound.

1. What is the difference between an infectious disease and an intoxication? Define toxemia.
2. Describe some general characteristics of exotoxins.
3. How do exotoxins get into host cells?
4. Describe the biological effects of several bacterial exotoxins.
5. Discuss the mechanisms by which exotoxins can damage cells.
6. What are the four types of exotoxins?
7. What is the mode of action of a leukocidin? Of a hemolysin?
8. Name two specific hemolysins.
9. What are the three main exotoxins have in human disease pathogenesis?

Endotoxins

Gram-negative bacteria have lipopolysaccharide (LPS) in the outer membrane of their cell wall that, under certain circumstances, is toxic to specific hosts. This LPS (*see figures 3.23–3.25*) is called an **endotoxin** because it is bound to the bacterium and is released when the microorganism lyses (**Box 34.1**). Some is also released during bacterial multiplication. The toxic component of the LPS is the lipid portion, called lipid A. Lipid A is not a single macromolecular structure but appears to be a complex array of lipid residues. The lipid A component exhibits all the properties (*see characteristic 5 on p. 801*) associated with endotoxicity and gram-negative bacteremia. [Gram-negative cell wall \(pp. 58–60\)](#)

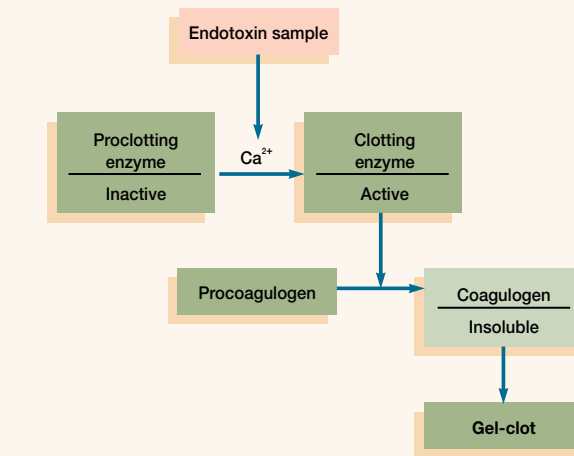
Box 34.1

Detection and Removal of Endotoxins

Bacterial endotoxins have plagued the pharmaceutical industry and medical device producers for years. For example, administration of drugs contaminated with endotoxins can result in complications—even death—to patients. Recently endotoxins have become a problem for individuals and firms working with cell cultures and genetic engineering. The result has been the development of sensitive tests and methods to identify and remove these endotoxins. The procedures must be very sensitive to trace amounts of endotoxins. Most firms have set a limit of 0.25 **endotoxin units** (E.U.), 0.025 ng/ml, or less as a release standard for their drugs, media, or products.

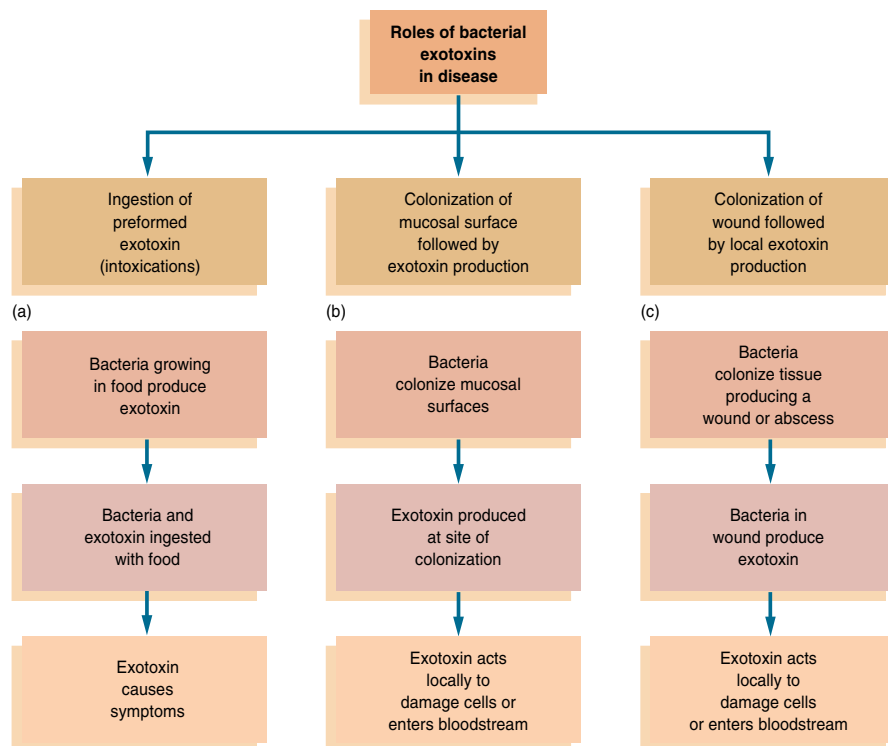
One of the most accurate tests for endotoxins is the *in vitro* *Limulus* amoebocyte lysate (LAL) assay. The assay is based on the observation that when an endotoxin contacts the clot protein from circulating amoebocytes of *Limulus*, a gel-clot forms. The assay kits available today contain calcium, proclotting enzyme, and procoagulogen. The proclotting enzyme is activated by bacterial endotoxin lipopolysaccharide and calcium to form active clotting enzyme (see **Box figure**). Active clotting enzyme then catalyzes the cleavage of procoagulogen into polypeptide subunits (coagulogen). The subunits join by disulfide bonds to form a gel-clot. Spectrophotometry is then used to measure the protein precipitated by the lysate. The LAL test is sensitive at the nanogram level but must be standardized against Food and Drug Administration Bureau of Biologics endotoxin reference standards. Results are reported in endotoxin units per milliliter and reference made to the particular reference standards used.

Removal of endotoxins presents more of a problem than their detection. Those present on glassware or medical devices can be inacti-



ated if the equipment is heated at 250°C for 30 minutes. Soluble endotoxins range in size from 20 kDa to large aggregates with diameters up to 0.1 μm. Thus they cannot be removed by conventional filtration systems. Manufacturers are currently developing special filtration systems and filtration cartridges that retain these endotoxins and help alleviate contamination problems.

Figure 34.7 Roles of Exotoxins in Disease Pathogenesis. Three ways (a, b, c) in which bacterial exotoxins can contribute to the progression of disease in a human.



Besides the preceding characteristics, bacterial endotoxins are

1. Heat stable
2. Toxic only at high doses (milligram per kilogram amounts)
3. Weakly immunogenic
4. Generally similar, despite source
5. Usually capable of producing general systematic effects: fever (are pyrogenic), shock, blood coagulation, weakness, diarrhea, inflammation, intestinal hemorrhage, and fibrinolysis (enzymatic breakdown of fibrin, the major protein component of blood clots)

The characteristics of endotoxins and exotoxins are contrasted in table 34.5.

The main biological effect of LPS is an indirect one, being mediated by host molecules and systems rather than by LPS directly. For example, endotoxins can initially activate Hageman Factor (blood clotting factor XII), which in turn activates up to four humoral systems: coagulation, complement, fibrinolytic, and kininogen systems.

Gram-negative endotoxins also indirectly induce a fever in the host by causing macrophages to release **endogenous pyrogens** that reset the hypothalamic thermostat. One important endogenous pyrogen is the cytokine interleukin-1 (*see figure 31.19*). Other cytokines released by macrophages, such as the tumor necrosis factor, also produce fever.

Recent evidence indicates that LPS affects macrophages and monocytes by binding to special plasma proteins called **LPS-binding proteins**. The LPS-LPS-binding protein complex then attaches to receptors on monocytes, macrophages, and other cells. This triggers several events, including the production of cytokines IL-1, IL-6, and tumor necrosis factor. As mentioned previously, IL-1 and tumor necrosis factor induce fever. These cytokines also promote other endotoxin effects: complement activation, coagulation, prostaglandin formation, and so forth.

-
1. Describe the chemical structure of the LPS endotoxin.
 2. List some general characteristics of endotoxins.
 3. How do gram-negative endotoxins induce fever in a mammalian host?
-

34.4 Microbial Mechanisms for Escaping Host Defenses

So far, we have discussed some of the ways viral and bacterial pathogens cause disease in a host. During the course of microbe and human evolution, these same pathogens have evolved ways for escaping host defenses. Many of these mechanisms are found throughout the microbial world and several are now discussed.

Evasion of Host Defenses by Viruses

As noted earlier in this chapter, the pathology arising from a viral infection is due to either (1) the host's immune response, which attacks virus-infected cells or produces hypersensitivity reactions (*see section 33.2*), or (2) the direct consequence of viral multipli-

cation within host cells. Viruses have evolved a variety of ways to suppress or evade the host's immune response. These mechanisms are just now becoming recognized through genomics and the functional analysis of specific gene products. Several examples follow.

Some viruses may mutate and change antigenic sites (antigenic drift, *see section 37.5*) on the virion proteins (the influenza virus) or may down-regulate the level of expression of viral cell surface proteins (the herpesvirus). Other viruses (HIV) may infect cells (T cells) of the immune system and diminish their function. HIV as well as the measles virus and cytomegalovirus cause the fusion of host cells. This allows these viruses to move from an infected cell to an uninfected cell without exposure to the antibody-containing fluids of the host. The herpesvirus may infect neurons that express little or no major histocompatibility complex molecules (*see section 32.4*). The adenovirus produces proteins that inhibit major histocompatibility complex function. Finally, hepatitis B virus infected cells produce large amounts of antigens not associated with the complete virus. These antigens bind the available neutralizing antibody (*see section 32.6*) so that there is insufficient free antibody to bind with the complete viral particle.

Evasion of Host Defenses by Bacteria

Bacteria also have evolved many mechanisms to evade host defenses. Because bacteria would not be well served either by the death of their host or their own death, their survival strategy is protection against host defenses rather than host destruction. Several of these evasive mechanisms are now discussed.

Evading the Complement System

To evade the activity of complement, some bacteria have capsules (*see chapter opening figure*) that prevent complement activation. Some gram-negative bacteria can lengthen the O chains in their lipopolysaccharide to prevent complement activation. Others such as *Neisseria gonorrhoeae* generate **serum resistance**. These bacteria have modified lipooligosaccharides on their surface that interfere with proper formation of the membrane attack complex (*see figure 31.14*) during the complement cascade. The virulent forms of *N. gonorrhoeae* that possess serum resistance are able to spread throughout the body of the host and cause systemic disease, whereas those *N. gonorrhoeae* that lack serum resistance remain localized in the genital tract.

[Complement activation \(pp. 714–18, 758–59\)](#)

Resisting Phagocytosis

As noted several times previously, before a phagocytic cell can phagocytose a bacterium, it must first directly contact the bacterium's surface. Some bacteria such as *Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae* can produce a slippery mucoid capsule that prevents the phagocyte from effectively contacting the bacterium. Other bacteria evade phagocytosis by producing specialized surface proteins such as the M protein on *S. pyogenes*. Like capsules, these proteins interfere with adherence between a phagocytic cell and the bacterium.

Bacterial pathogens can resist phagocytosis in quite different ways. For example, *Staphylococcus* produces leukocidins